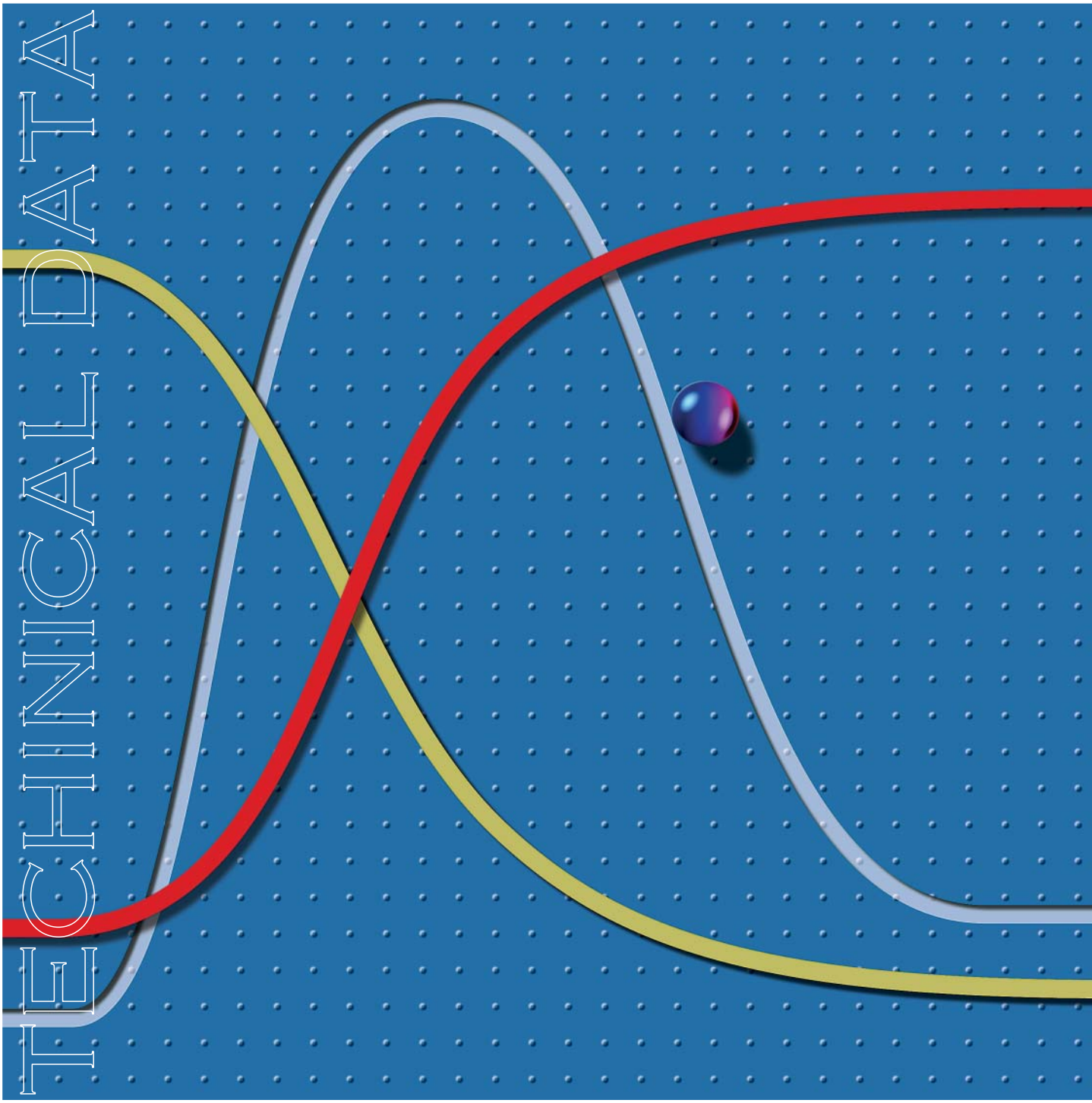
Super Core™

Magnetic Property Curves

JNEX-CORE JNHF-CORE

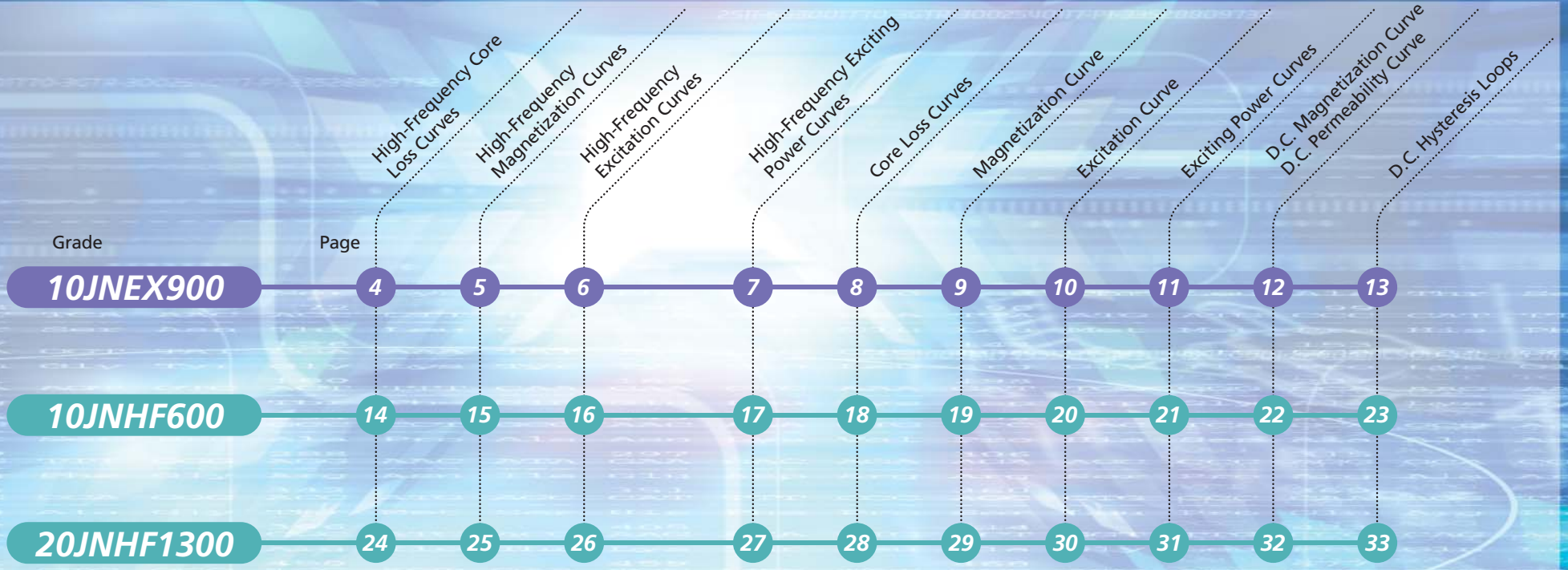


JFE Steel Corporation

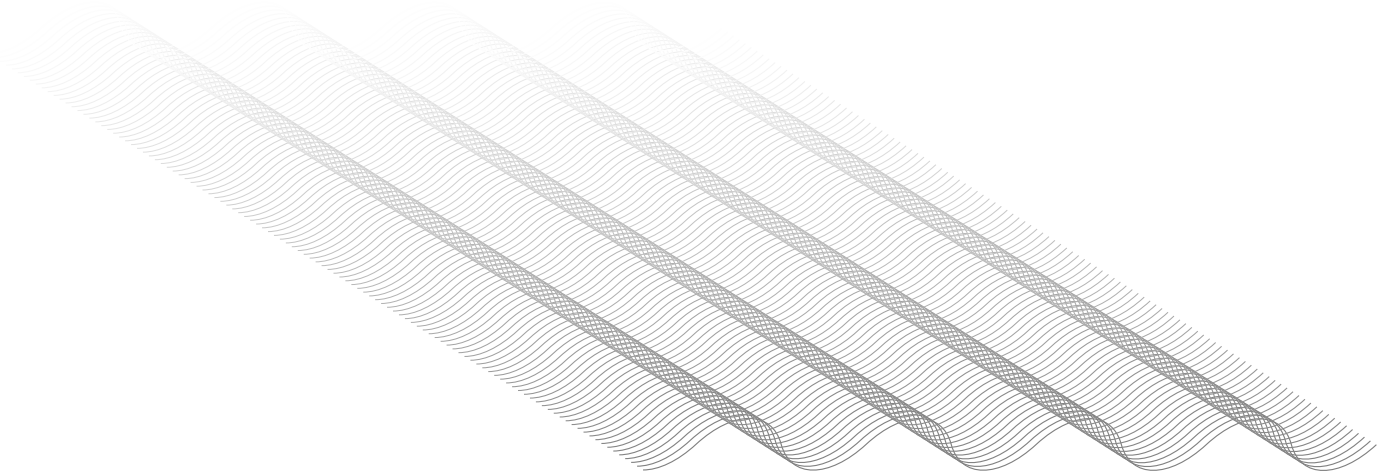
JFE Steel's Super Core™

Magnetic property curves

Typical magnetic property curves for each grade of Super Core™ is shown on the pages listed in the tables below.



•Super Core is a registered trademark or trademark of JFE Steel Corporation in the United States and other countries.



Unit Conversion Factors

Magnetizing Force

	Oersted	A/m	A/in
Oersted	1	7.96×10	2.02
A/m	1.256×10^{-2}	1	2.54×10^{-2}
A/in	4.95×10^{-1}	3.94×10	1

Core Loss

	w/kg	w/lb
w/kg	1	4.54×10^{-1}
w/lb	2.204	1

Magnetic Flux Density

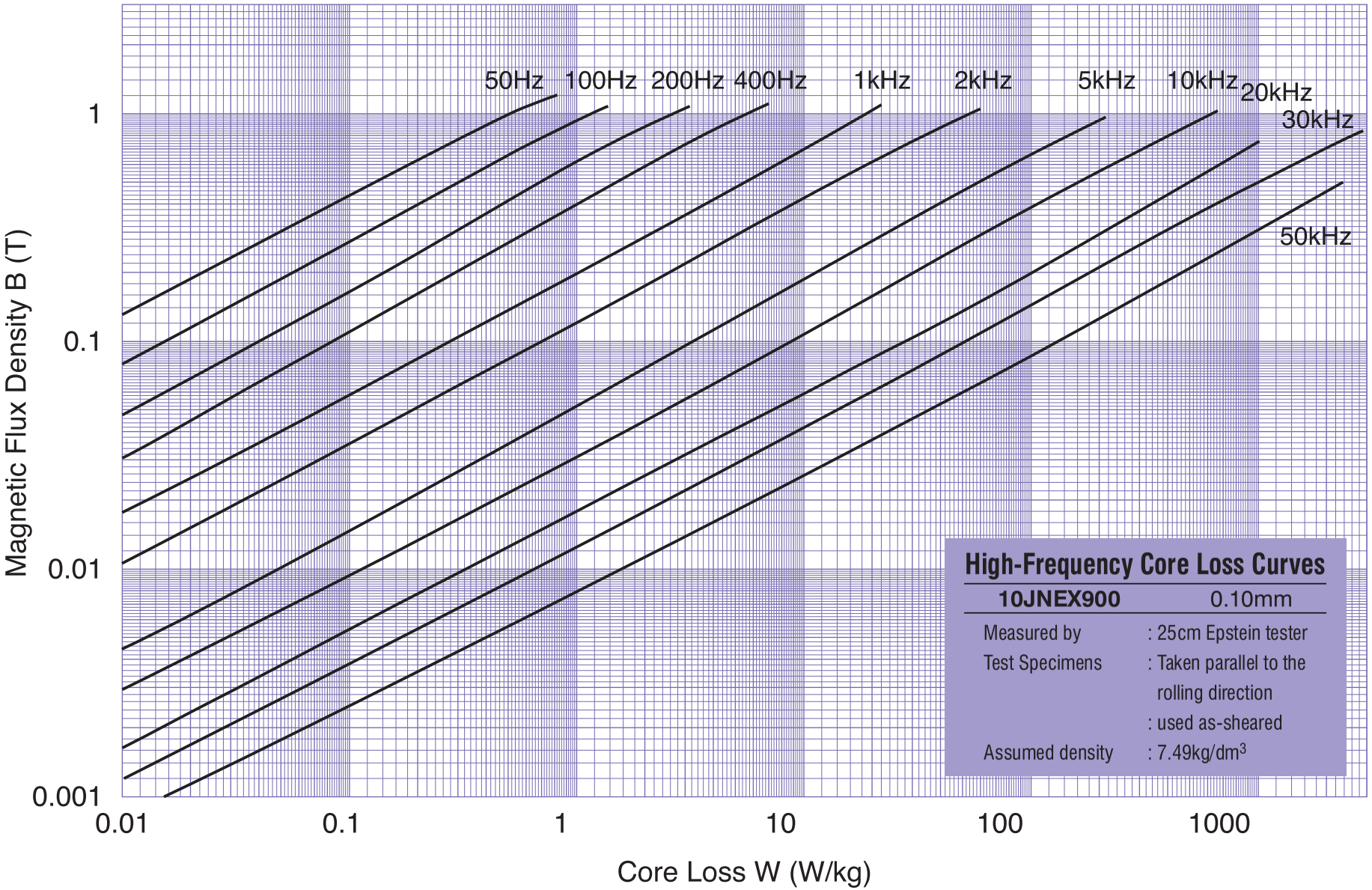
	Gauss	Tesla	Line/in^2	Wb/m^2
Gauss	1	10^{-4}	6.45	10^{-4}
Tesla	10^4	1	6.45×10^4	1
Line/in^2	1.55×10^{-1}	1.55×10^{-5}	1	1.55×10^{-5}
Wb/m^2	10^4	1	6.45×10^4	1

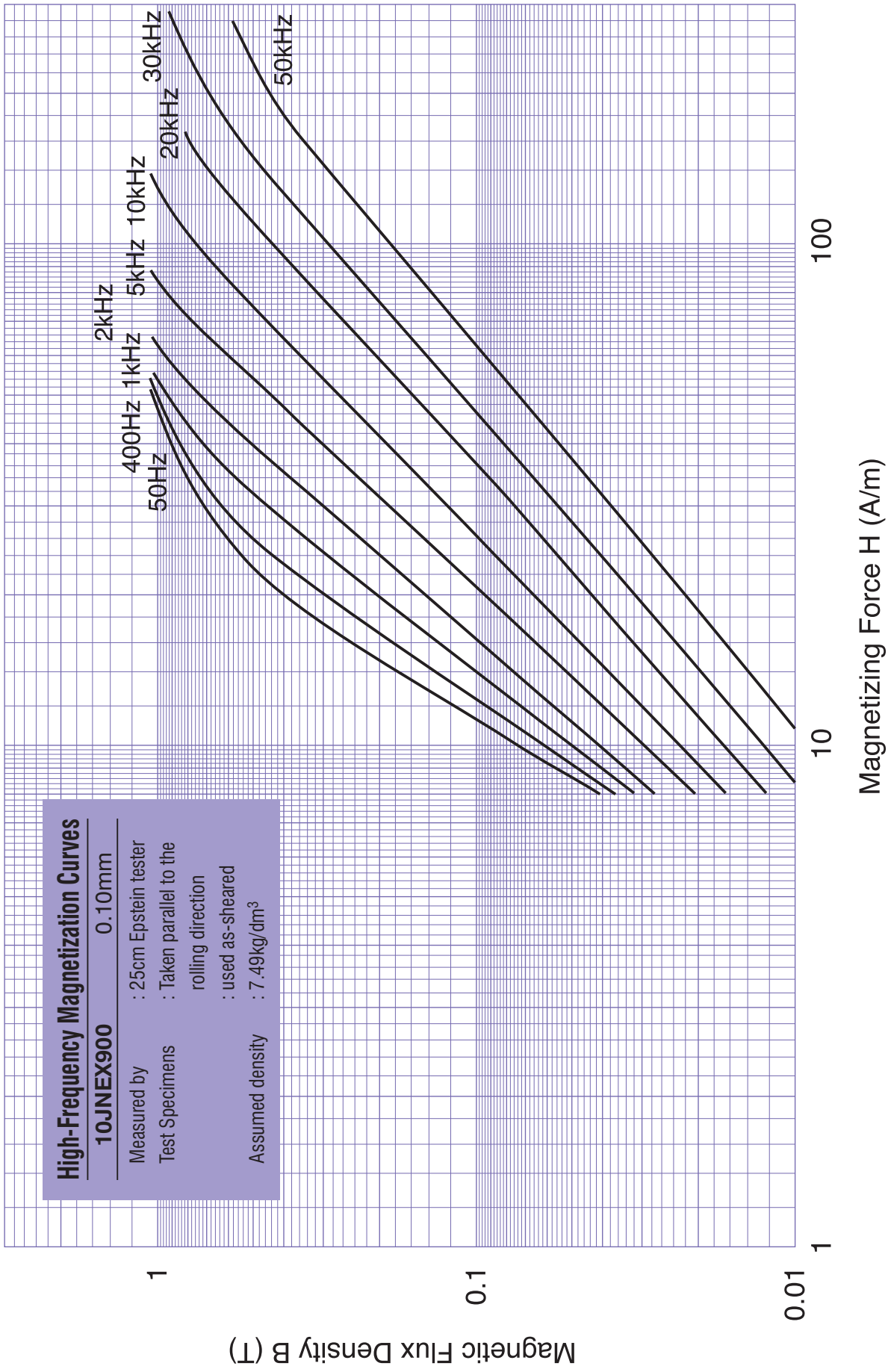
Magnetic Permeability

	Gauss/Oe	H/m
CGS Electro-magnetic Unit	1	1.257×10^{-6}
Henry per Meter(H/m)	7.958×10^5	1

Conversion of Core Loss as per AISA

	W/kg 60Hz	W/kg 50Hz
W/kg 60Hz	1	0.79
W/kg 50Hz	1.266	1

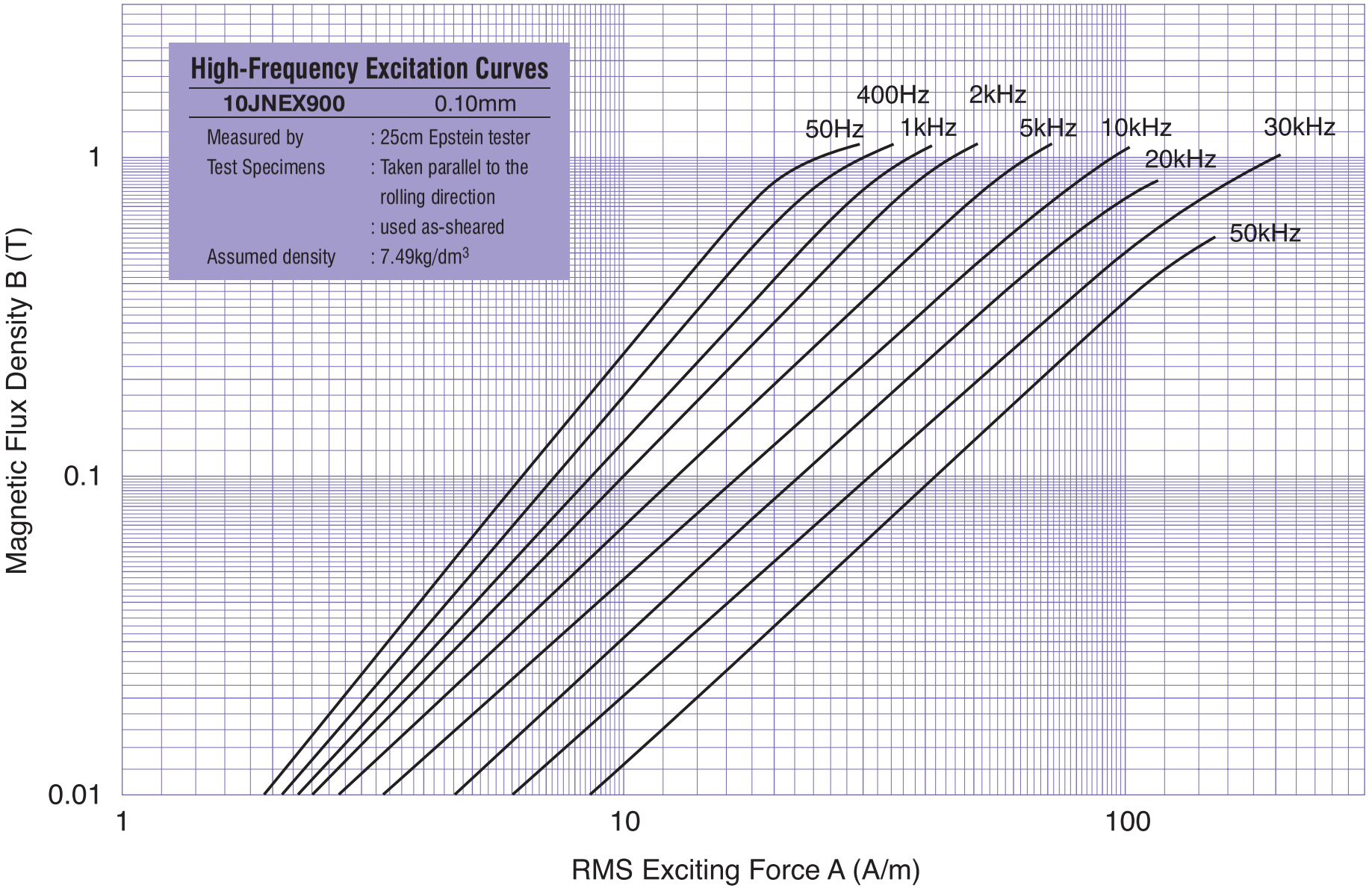

50Hz
100Hz
200Hz
400Hz
1kHz
2kHz
5kHz
10kHz
20kHz
30kHz
50kHz
Magnetic Flux Density B (T)
1
0.1
0.01
0.001
0.01
0.1
1
10
100
1000
Core Loss W (W/kg)
High-Frequency Core Loss Curves
10JNEX900 0.10mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.49kg/dm3

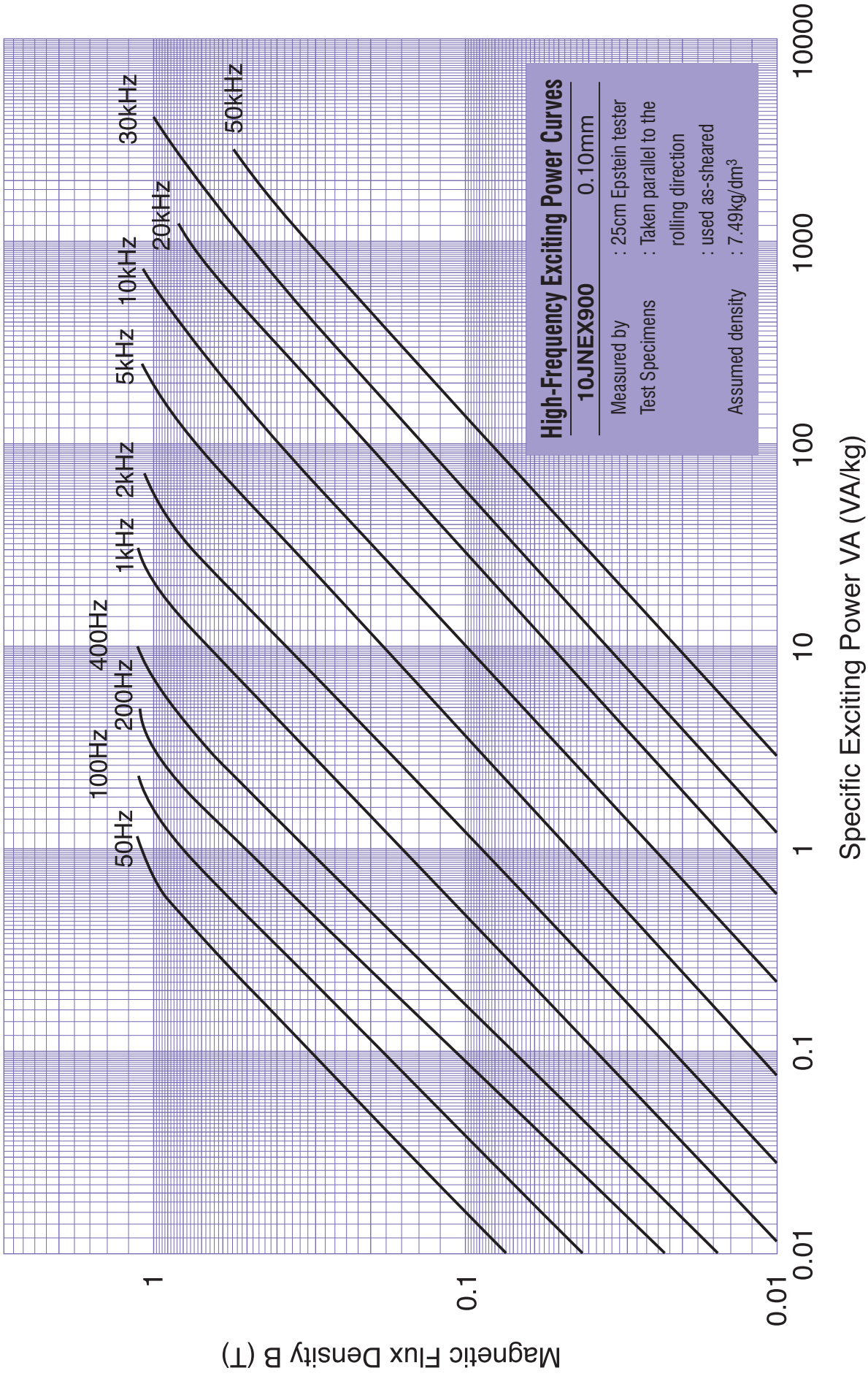

High-Frequency Magnetization Curves
10JNEX900 0.10mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.49kg/dm3
50Hz
400Hz
1kHz
2kHz
5kHz
10kHz
20kHz
30kHz
50kHz
Magnetic Flux Density B (T)
1
0.1
0.01
1
10
100
Magnetizing Force H (A/m)

High-Frequency Excitation Curves

10JNEX900 0.10mm

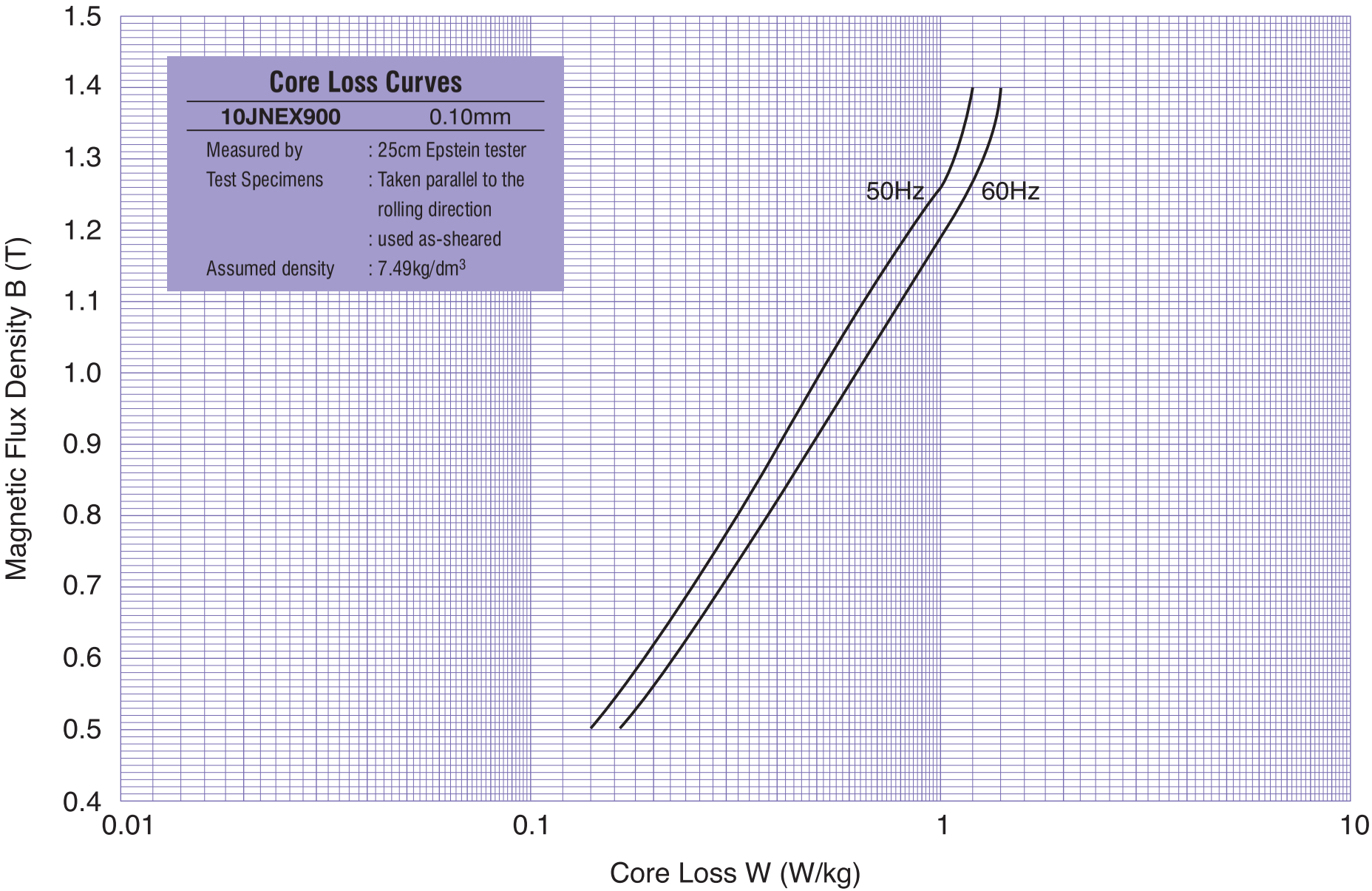
Measured by	: 25cm Epstein tester
Test Specimens	: Taken parallel to the rolling direction
	: used as-sheared
Assumed density	: 7.49kg/dm^3

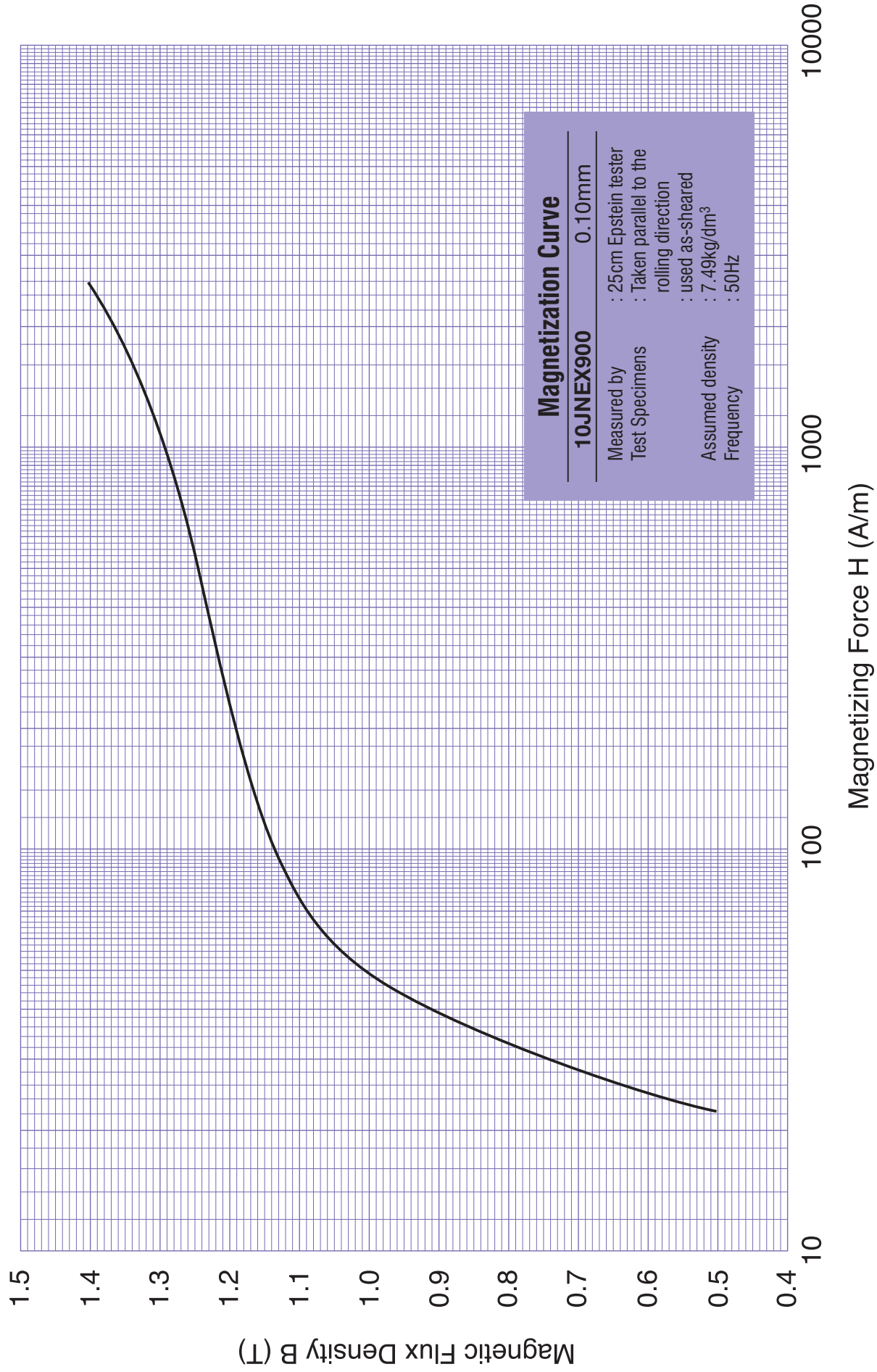


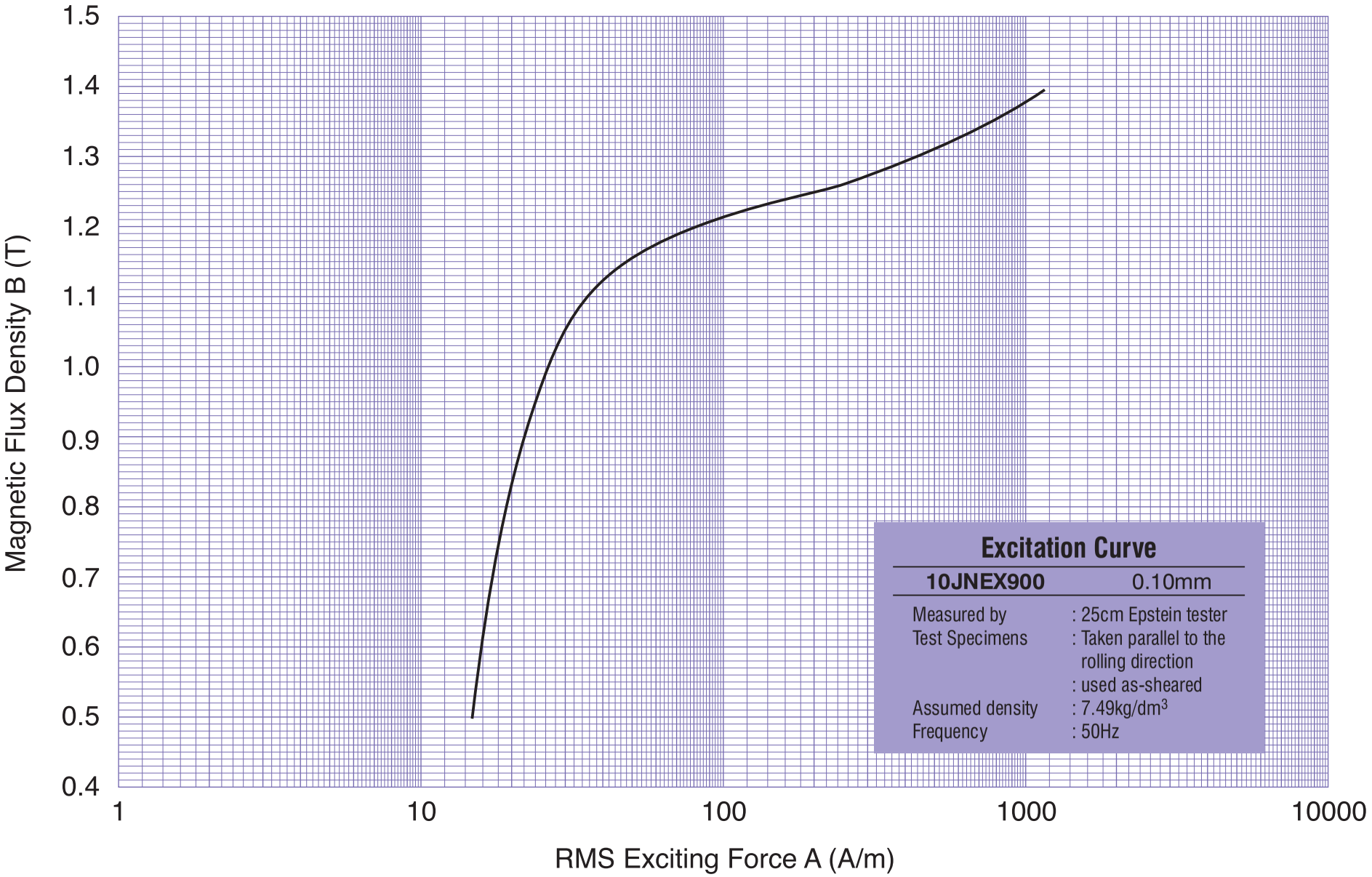

High-Frequency Exciting Power Curves
10JNEX900 0.10mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.49kg/dm3
50Hz
100Hz
200Hz
400Hz
1kHz
2kHz
5kHz
10kHz
20kHz
30kHz
50kHz
Magnetic Flux Density B (T)
1
0.1
0.01
0.01
0.1
1
10
100
1000
10000
Specific Exciting Power VA (VA/kg)

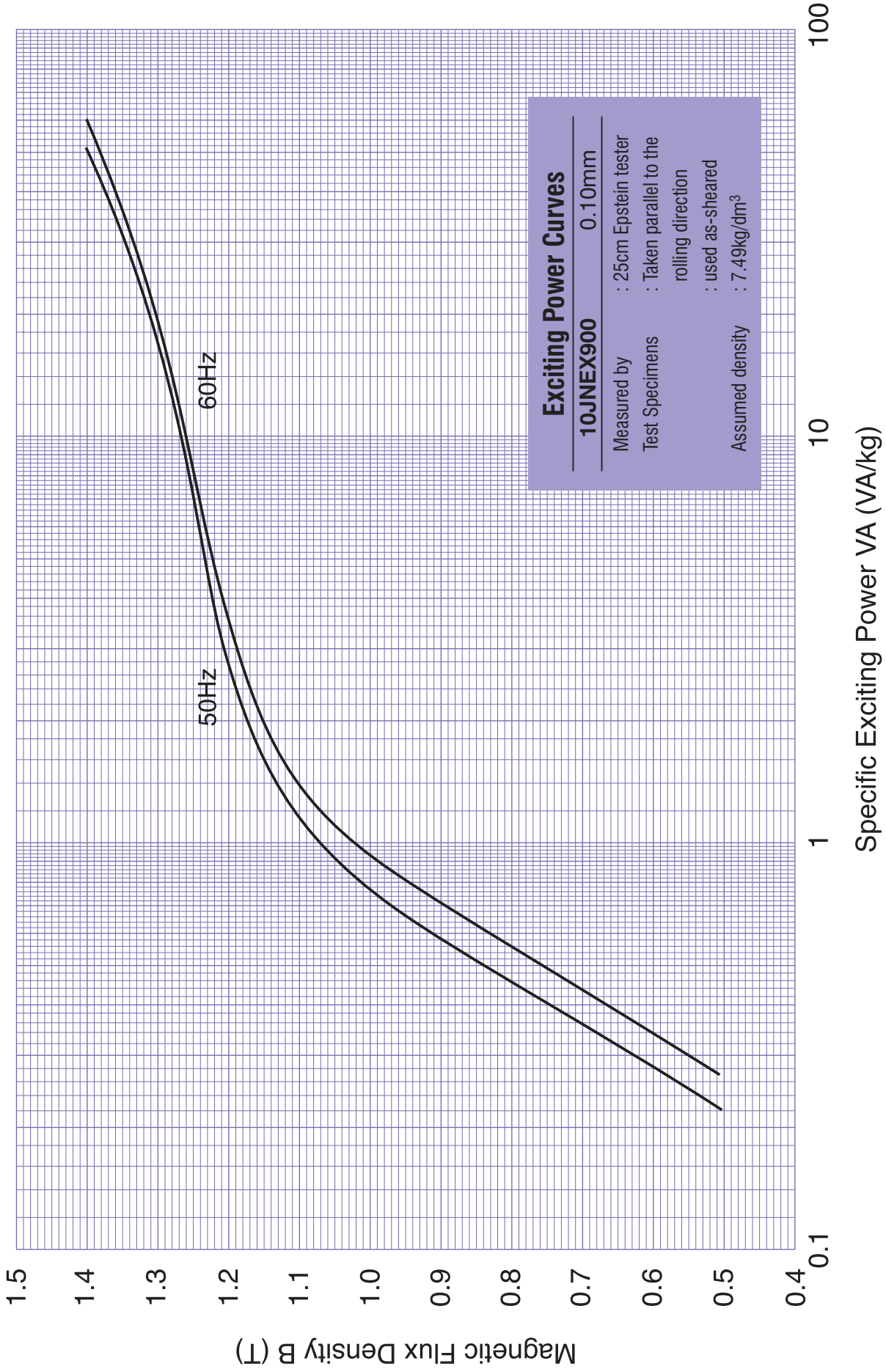
Core Loss Curves

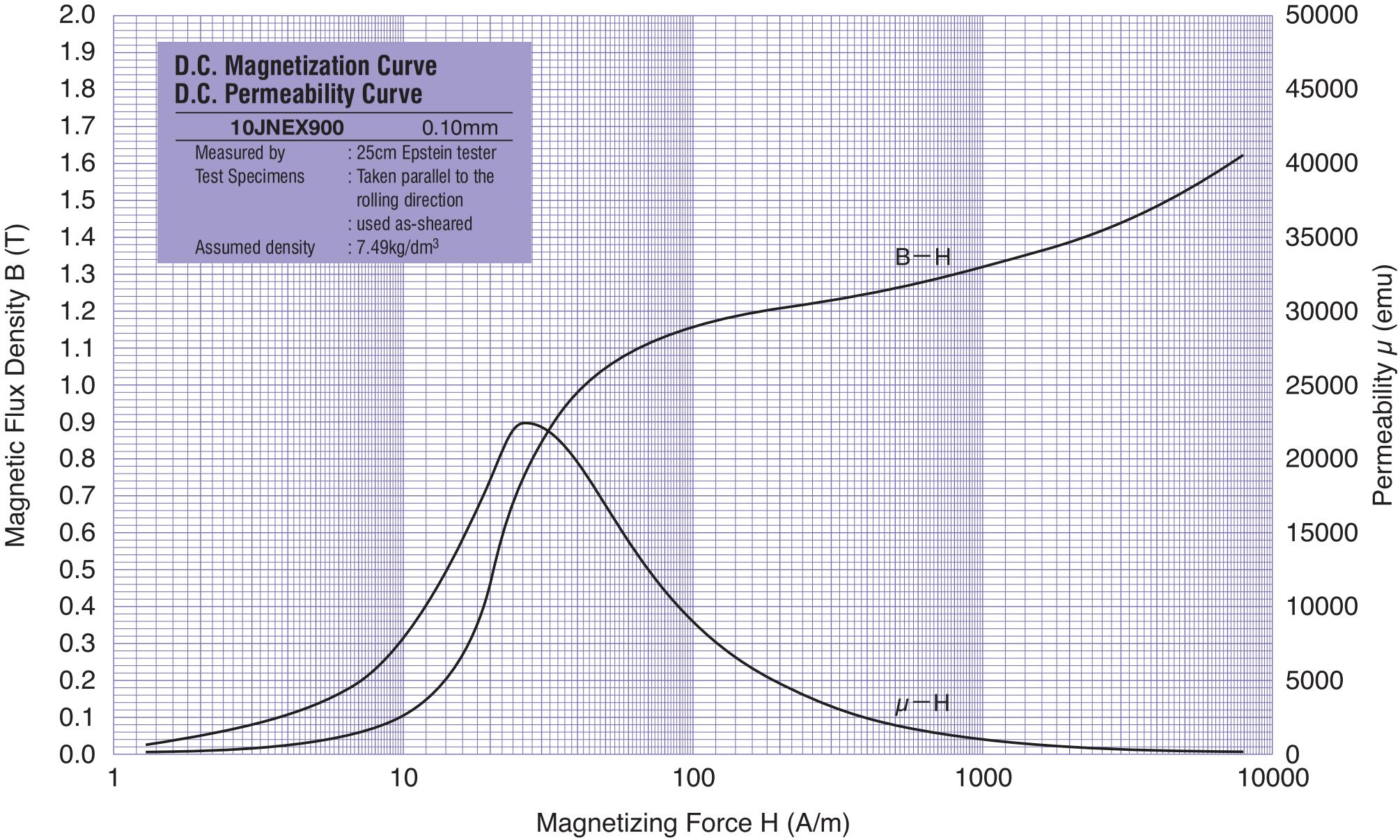
10JNEX900	0.10mm
Measured by	: 25cm Epstein tester
Test Specimens	: Taken parallel to the rolling direction
	: used as-sheared
Assumed density	: 7.49kg/dm^3

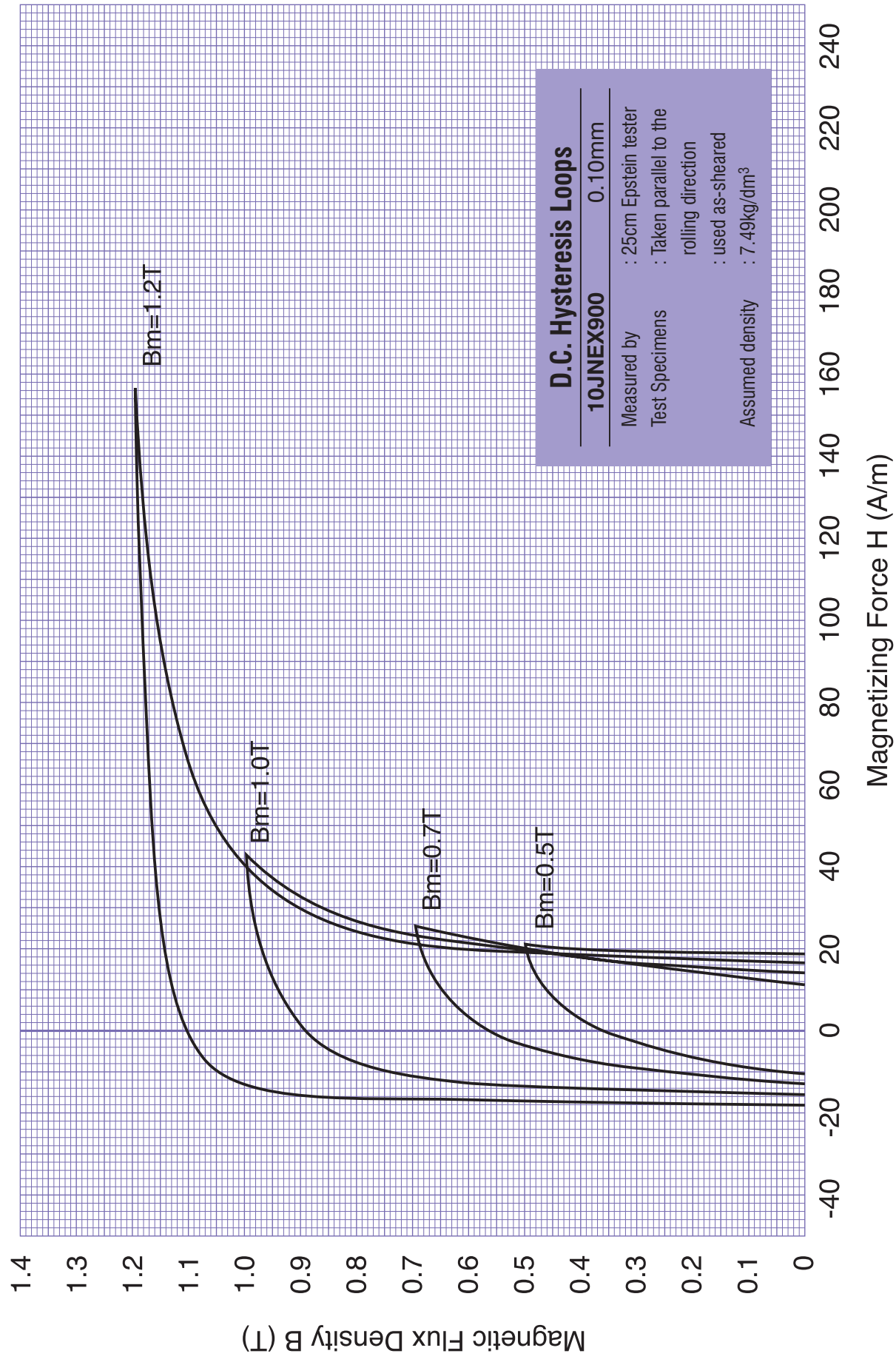


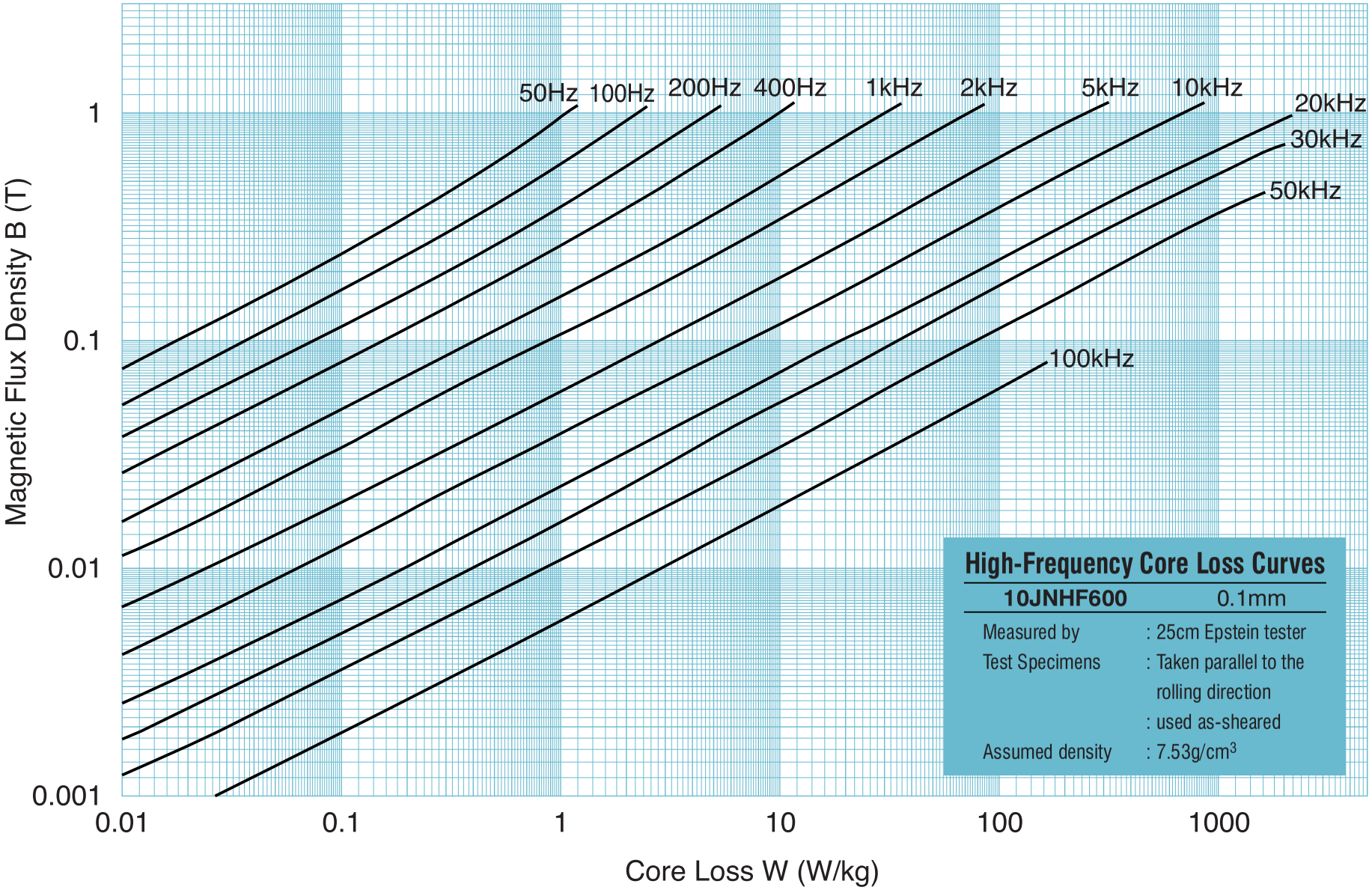

Magnetization Curve
10JNEX900 0.10mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.49kg/dm3
Frequency : 50Hz
Magnetic Flux Density B (T)
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
10
100
1000
10000
Magnetizing Force H (A/m)


Magnetic Flux Density B (T)
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1
10
100
1000
10000
RMS Exciting Force A (A/m)
Excitation Curve
10JNEX900 0.10mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.49kg/dm3
Frequency : 50Hz


Exciting Power Curves
10JNEX900 0.10mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.49kg/dm3
60Hz
50Hz
Magnetic Flux Density B (T)
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.1
1
10
100
Specific Exciting Power VA (VA/kg)

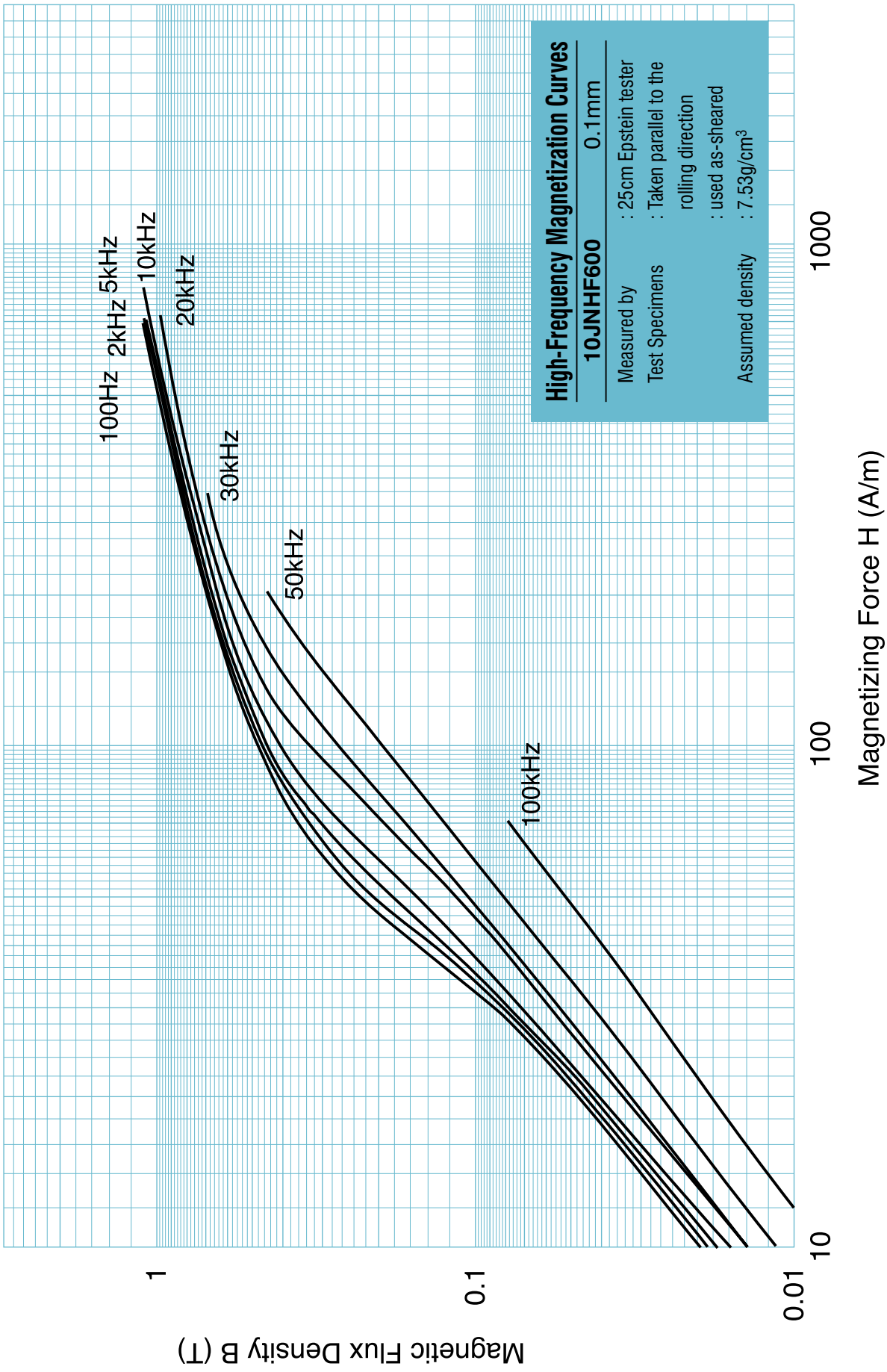

D.C. Magnetization Curve
D.C. Permeability Curve
10JNEX900 0.10mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.49kg/dm3
B−H
μ−H
Magnetic Flux Density B (T)
Permeability μ (emu)
Magnetizing Force H (A/m)
2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
50000
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
1
10
100
1000
10000

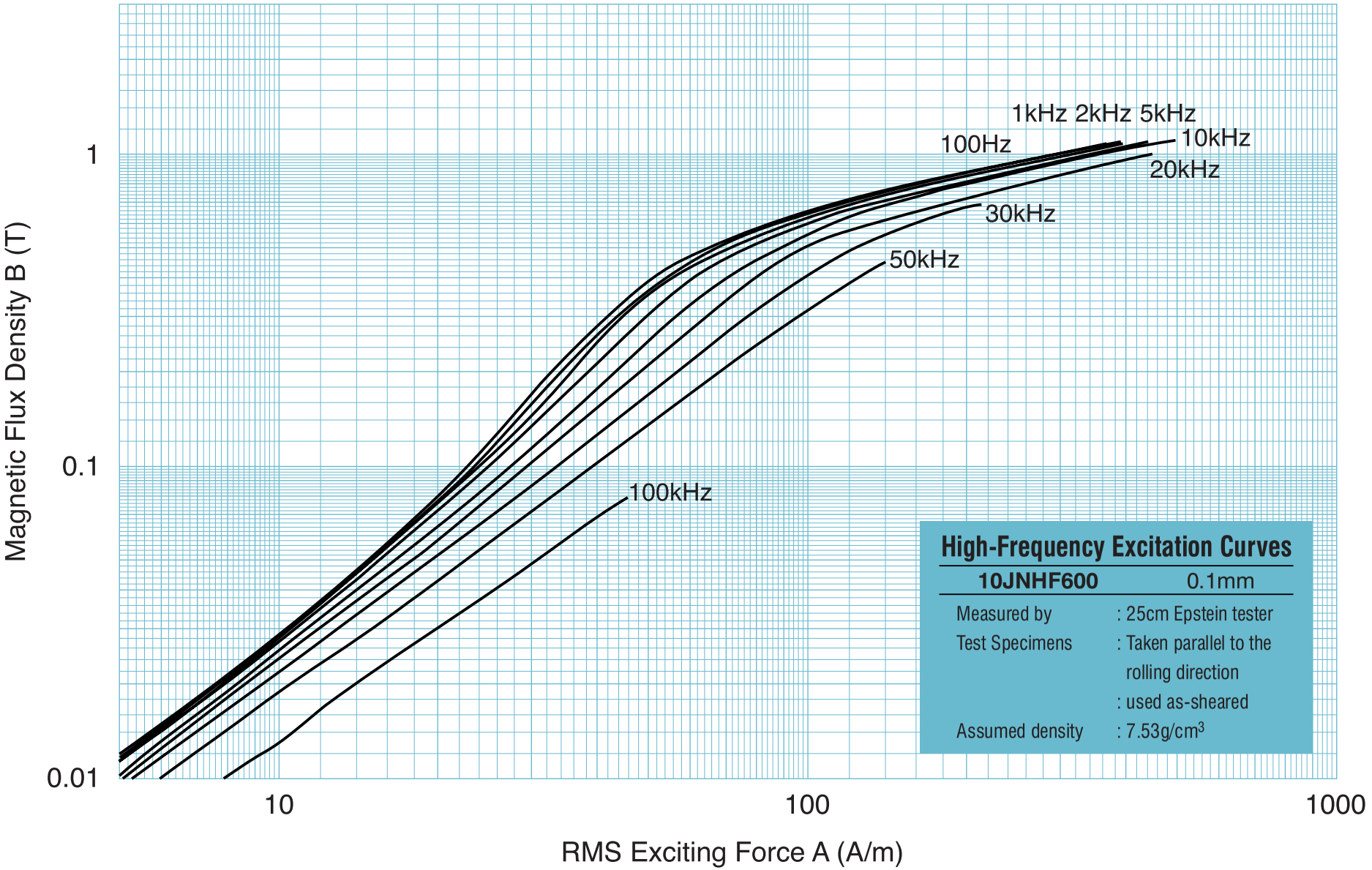

D.C. Hysteresis Loops
10JNEX900 0.10mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.49kg/dm3
Bm=1.2T
Bm=1.0T
Bm=0.7T
Bm=0.5T
Magnetic Flux Density B (T)
Magnetizing Force H (A/m)
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-40
-20
0
20
40
60
80
100
120
140
160
180
200
220
240



High-Frequency Core Loss Curves

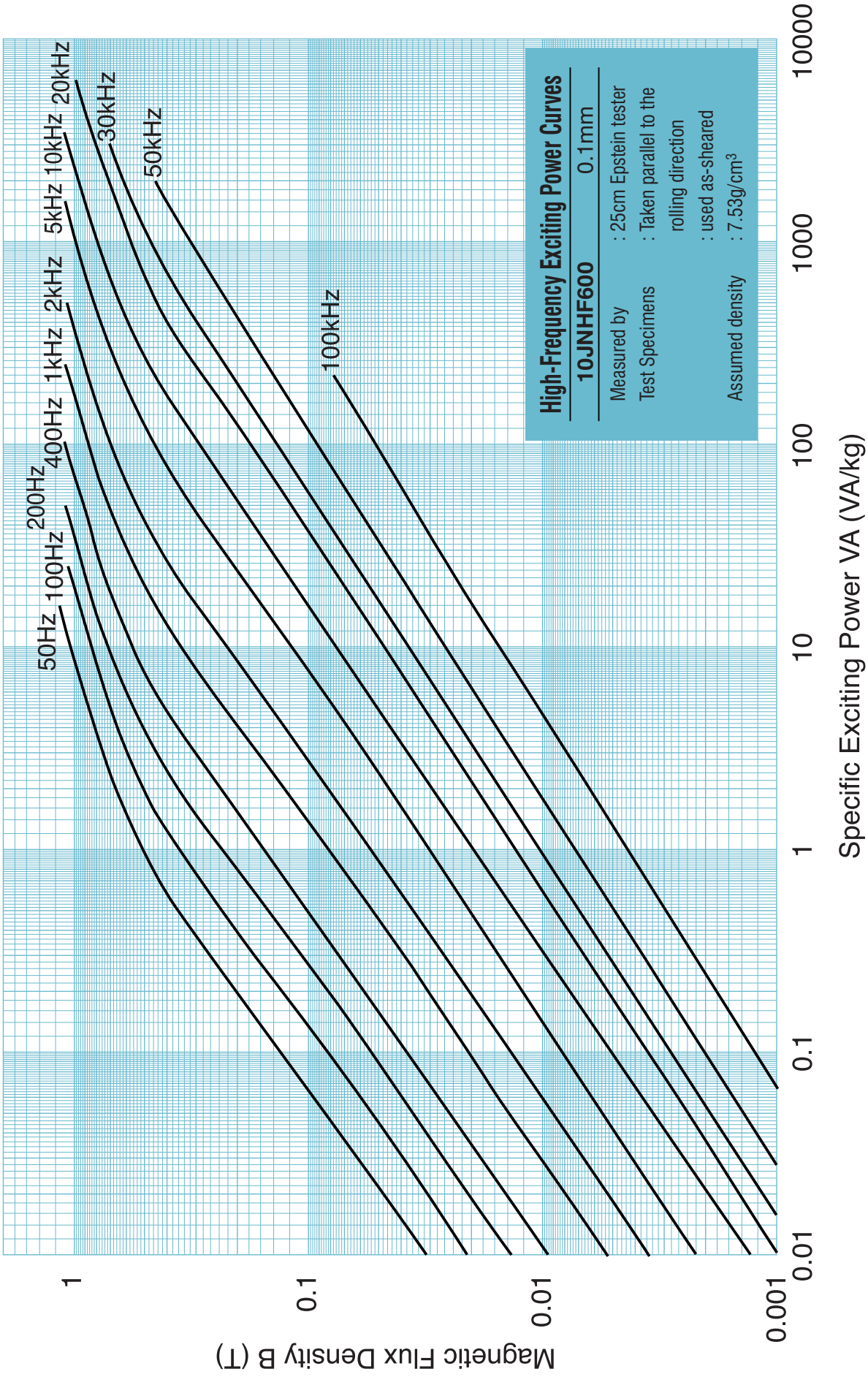
10JNHF600	0.1mm
Measured by	: 25cm Epstein tester
Test Specimens	: Taken parallel to the rolling direction
	: used as-sheared
Assumed density	: 7.53g/cm^3

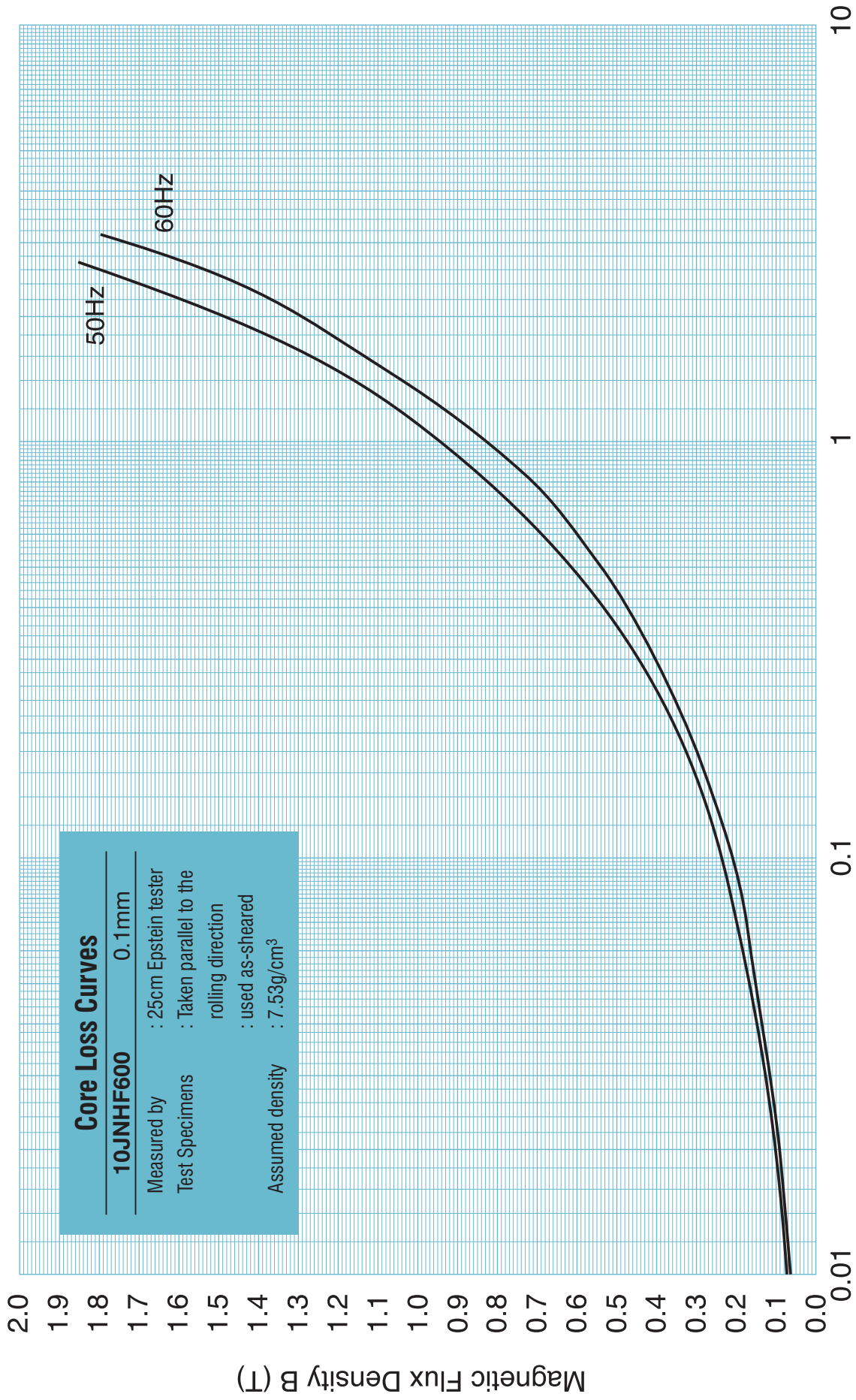

High-Frequency Magnetization Curves
10JNHF600 0.1mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.53g/cm3
100Hz
2kHz
5kHz
10kHz
20kHz
30kHz
50kHz
100kHz
1
0.1
0.01
10
100
1000
Magnetic Flux Density B (T)
Magnetizing Force H (A/m)

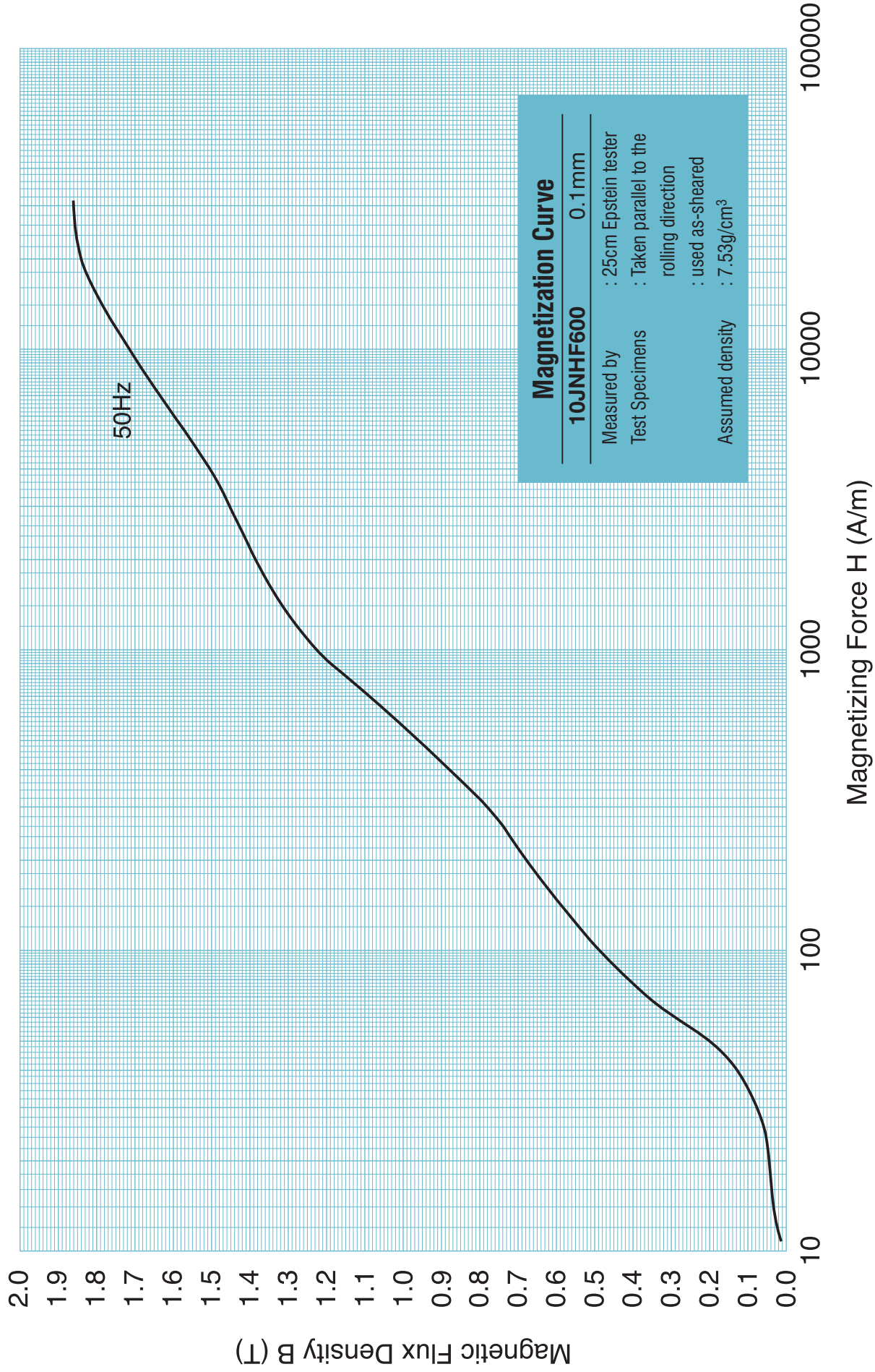


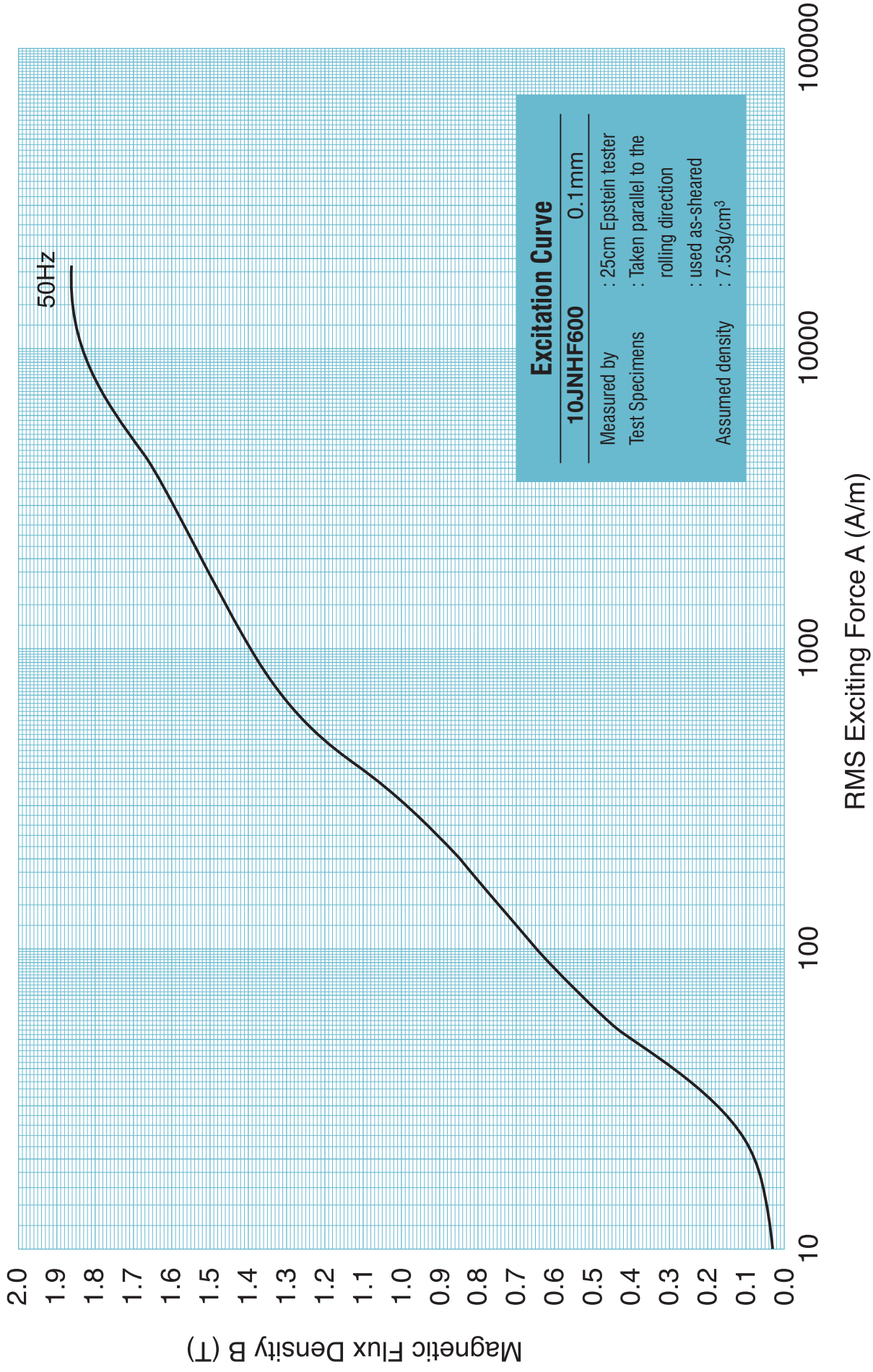
High-Frequency Excitation Curves

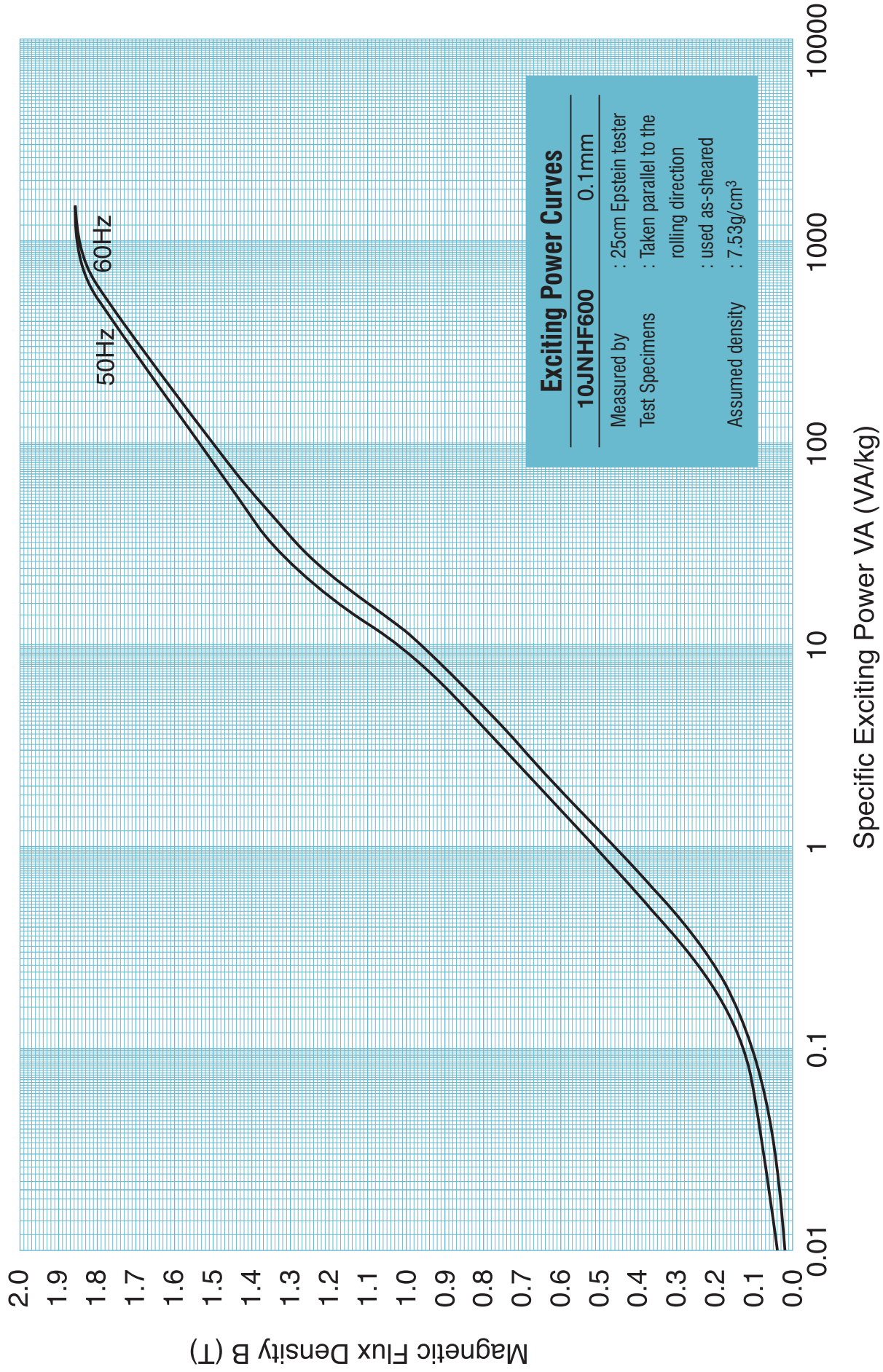
10JNHF600	0.1mm
Measured by	: 25cm Epstein tester
Test Specimens	: Taken parallel to the rolling direction
	: used as-sheared
Assumed density	: 7.53g/cm^3

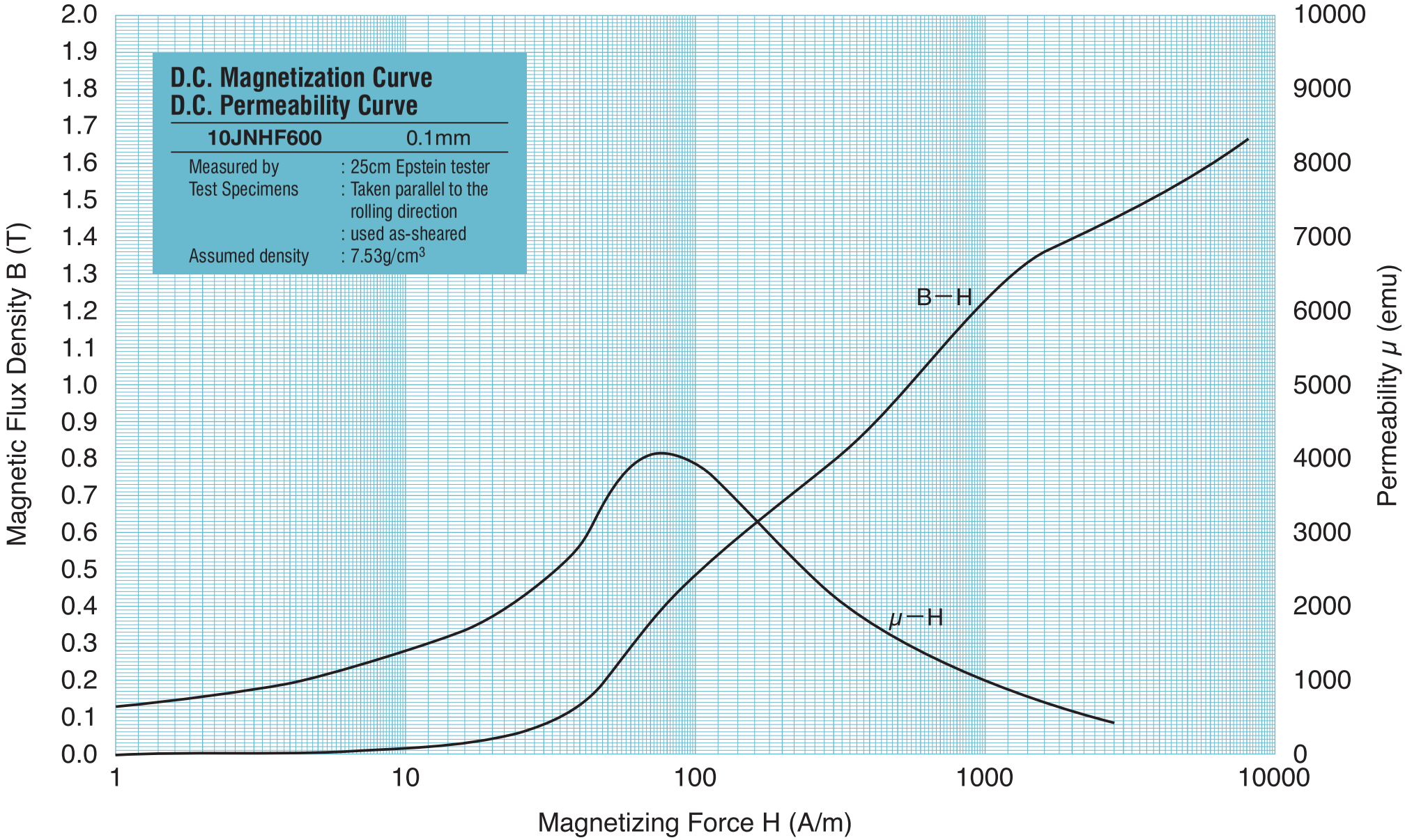

High-Frequency Exciting Power Curves
10JNHF600 0.1mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.53g/cm3
50Hz
100Hz
200Hz
400Hz
1kHz
2kHz
5kHz
10kHz
20kHz
30kHz
50kHz
100kHz
Magnetic Flux Density B (T)
1
0.1
0.01
0.001
0.01
0.1
1
10
100
1000
10000
Specific Exciting Power VA (VA/kg)

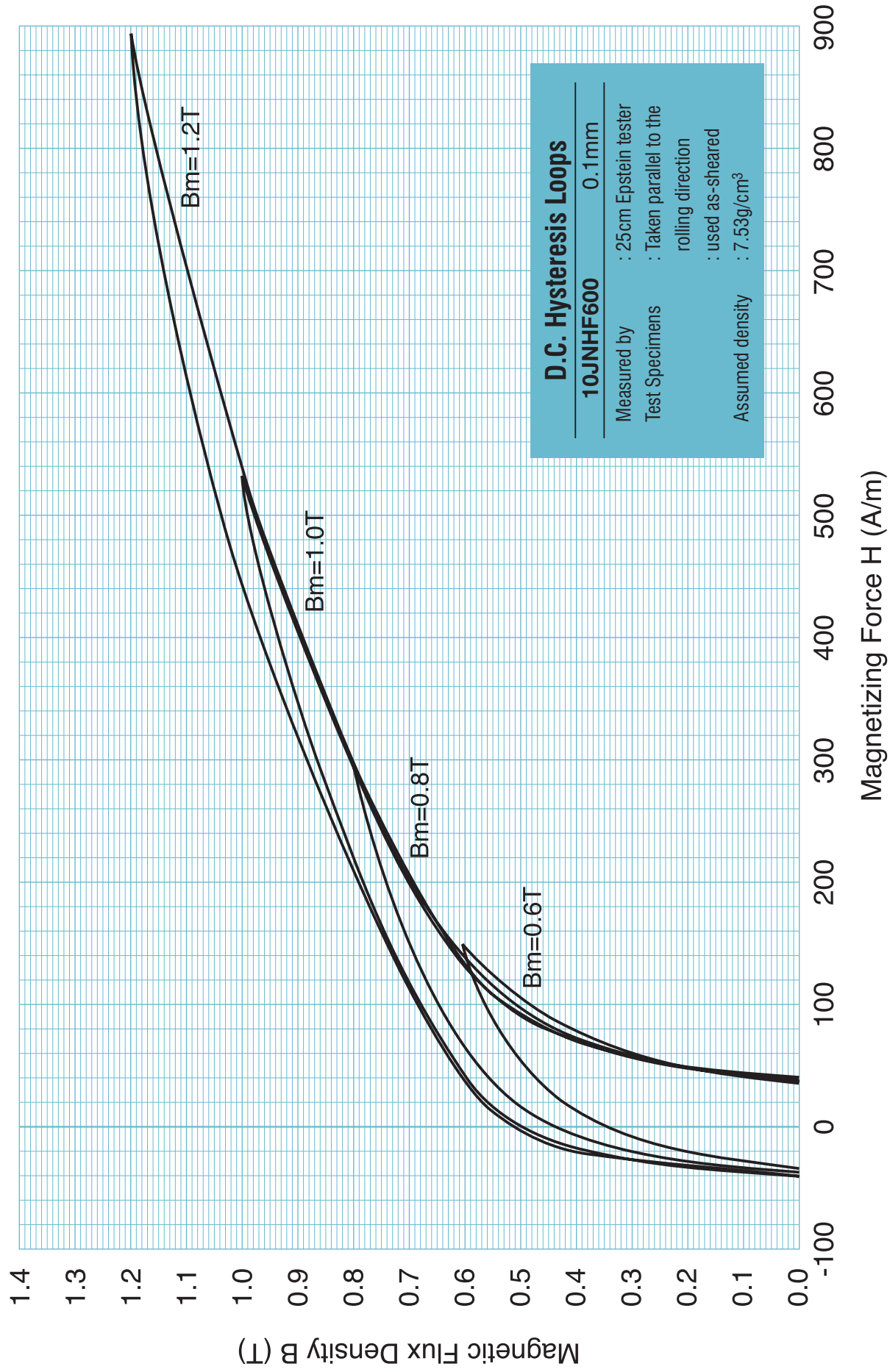

Core Loss Curves
10JNHF600 0.1mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.53g/cm3
50Hz
60Hz
Magnetic Flux Density B (T)
Core Loss W (W/kg)
2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.01
0.1
1
10

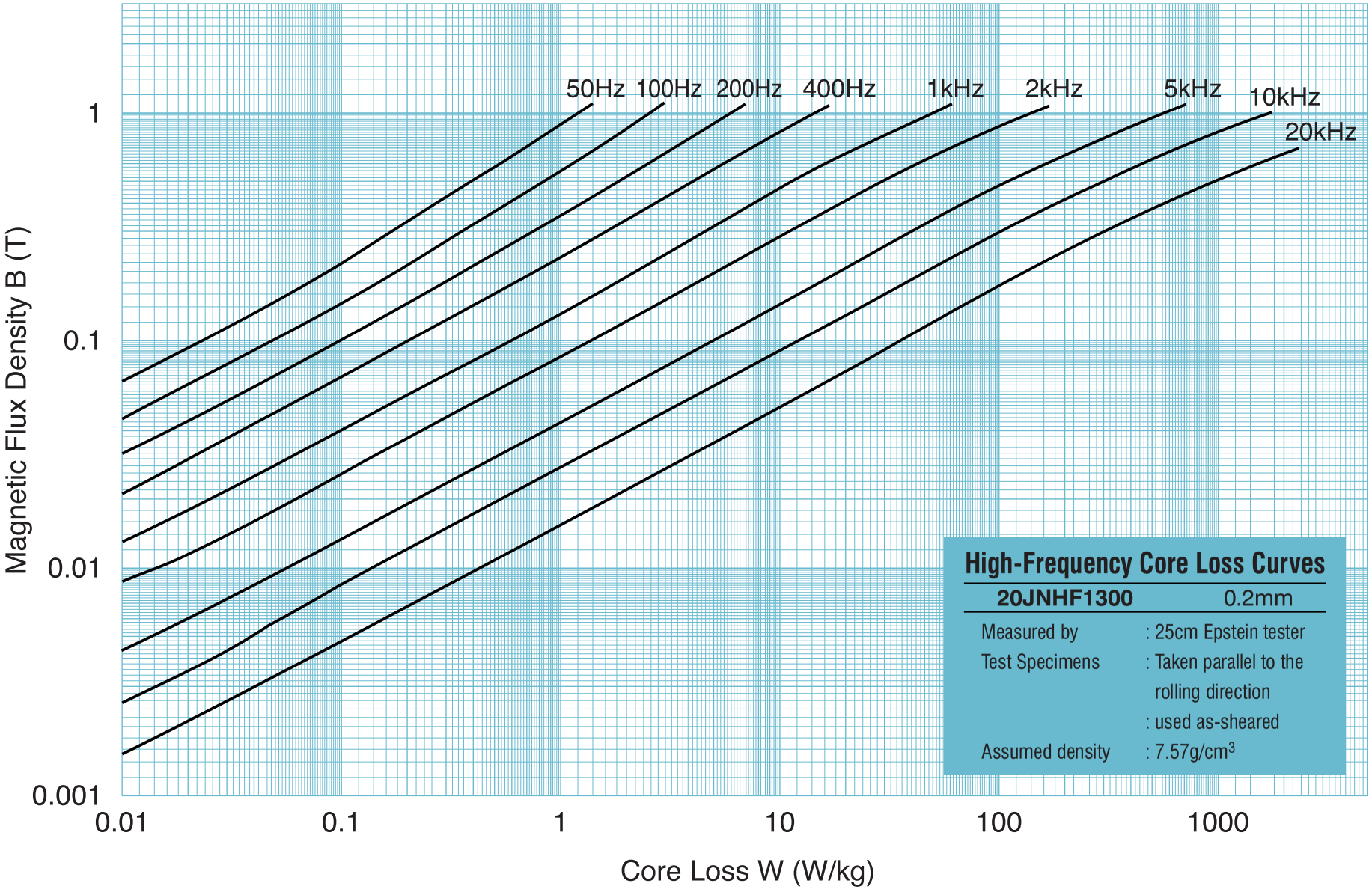

Magnetic Flux Density B (T)
2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
10
100
1000
10000
100000
Magnetizing Force H (A/m)
50Hz
Magnetization Curve
10JNHF600
0.1mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.53g/cm3

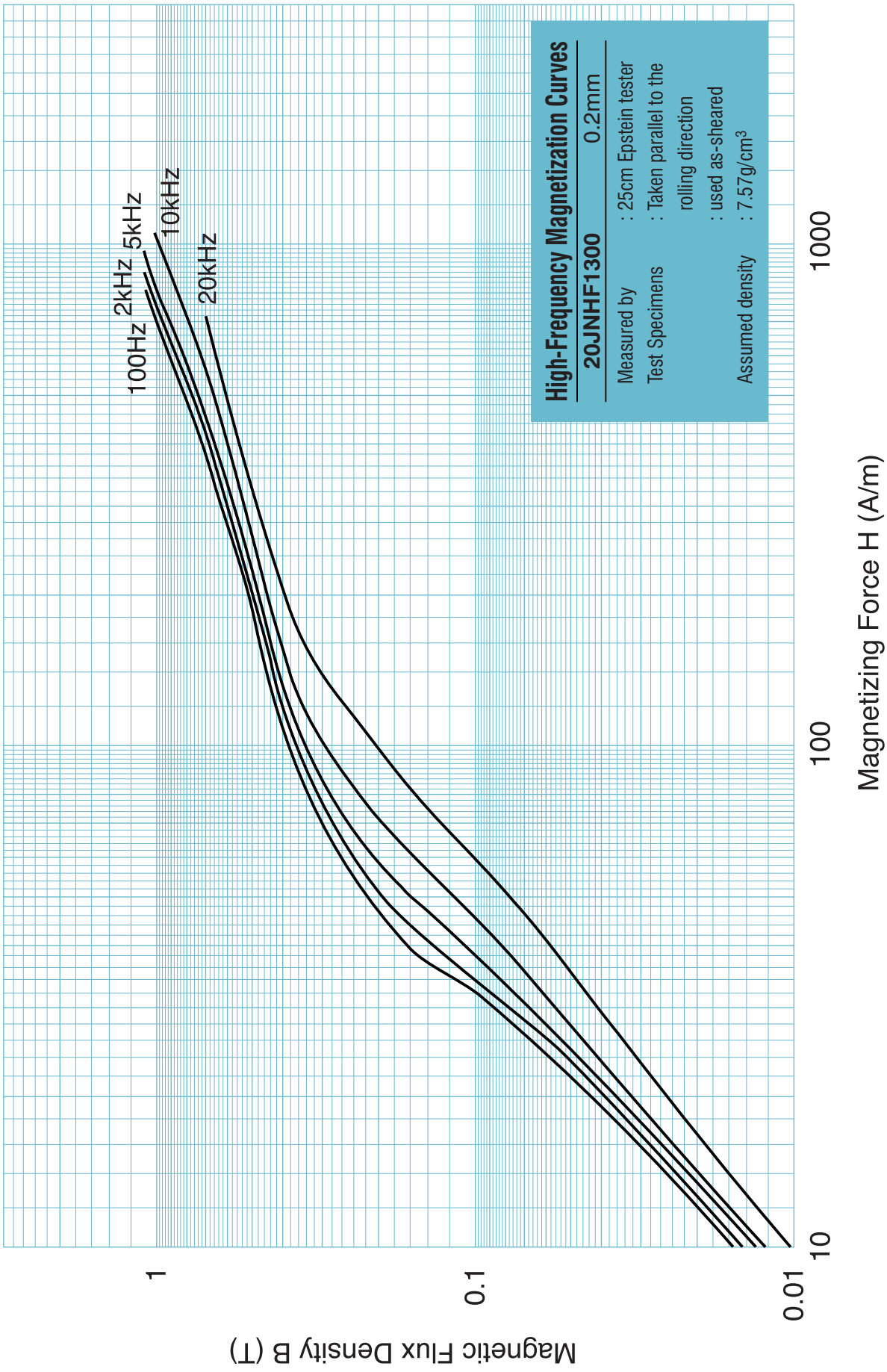

Excitation Curve
10JNHF600 0.1mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.53g/cm³
50Hz
Magnetic Flux Density B (T)
RMS Exciting Force A (A/m)
2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
10
100
1000
10000
100000

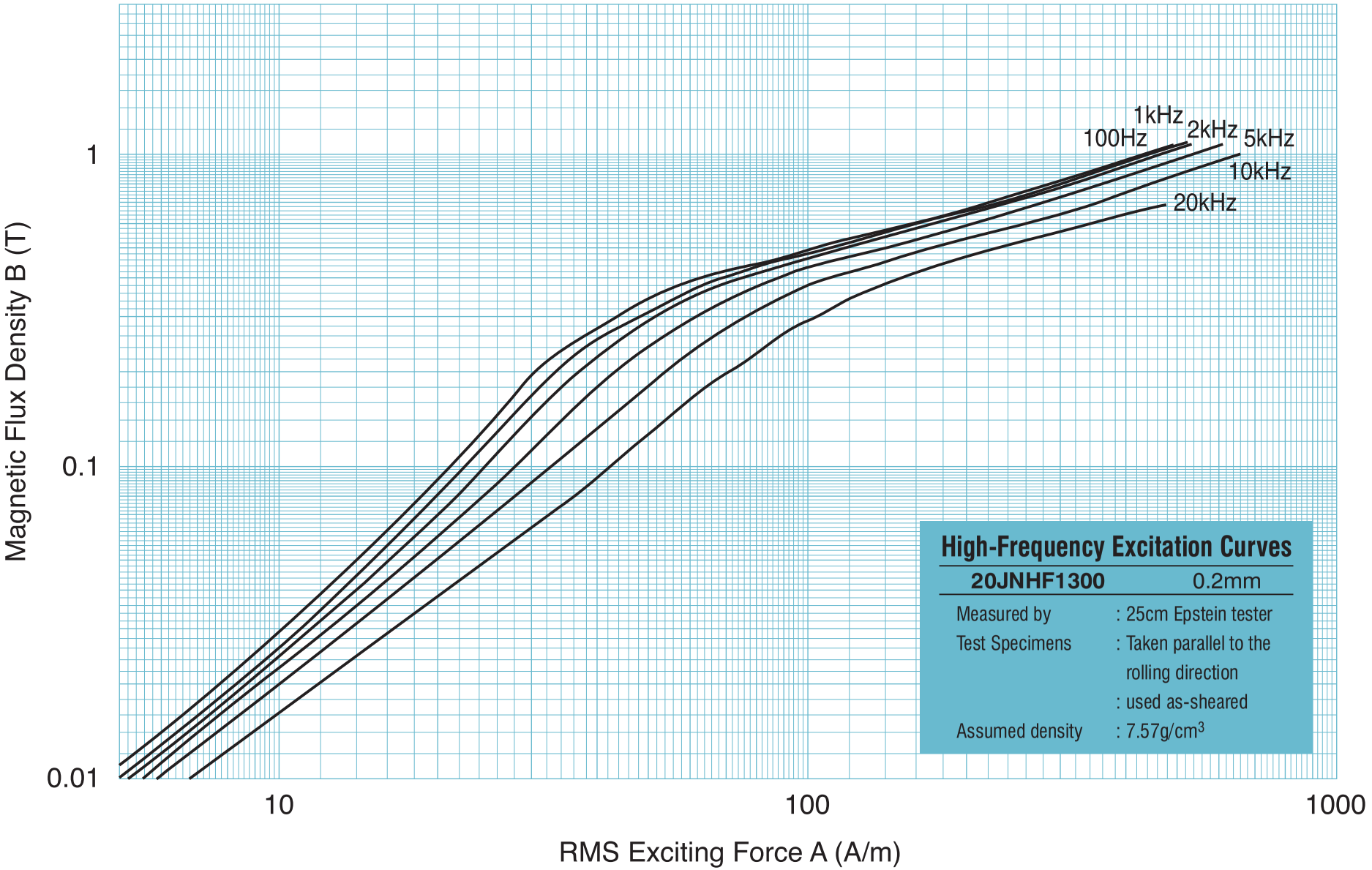

Exciting Power Curves
10JNHF600 0.1mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.53g/cm3
50Hz
60Hz
Magnetic Flux Density B (T)
Specific Exciting Power VA (VA/kg)
2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.01
0.1
1
10
100
1000
10000


D.C. Magnetization Curve
D.C. Permeability Curve
10JNHF600 0.1mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.53g/cm3
B−H
μ−H
Magnetic Flux Density B (T)
Permeability μ (emu)
Magnetizing Force H (A/m)
2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
10000
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
1
10
100
1000
10000


D.C. Hysteresis Loops
10JNHF600 0.1mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.53g/cm³
Bm=1.2T
Bm=1.0T
Bm=0.8T
Bm=0.6T
Magnetic Flux Density B (T)
Magnetizing Force H (A/m)
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
-100
0
100
200
300
400
500
600
700
800
900

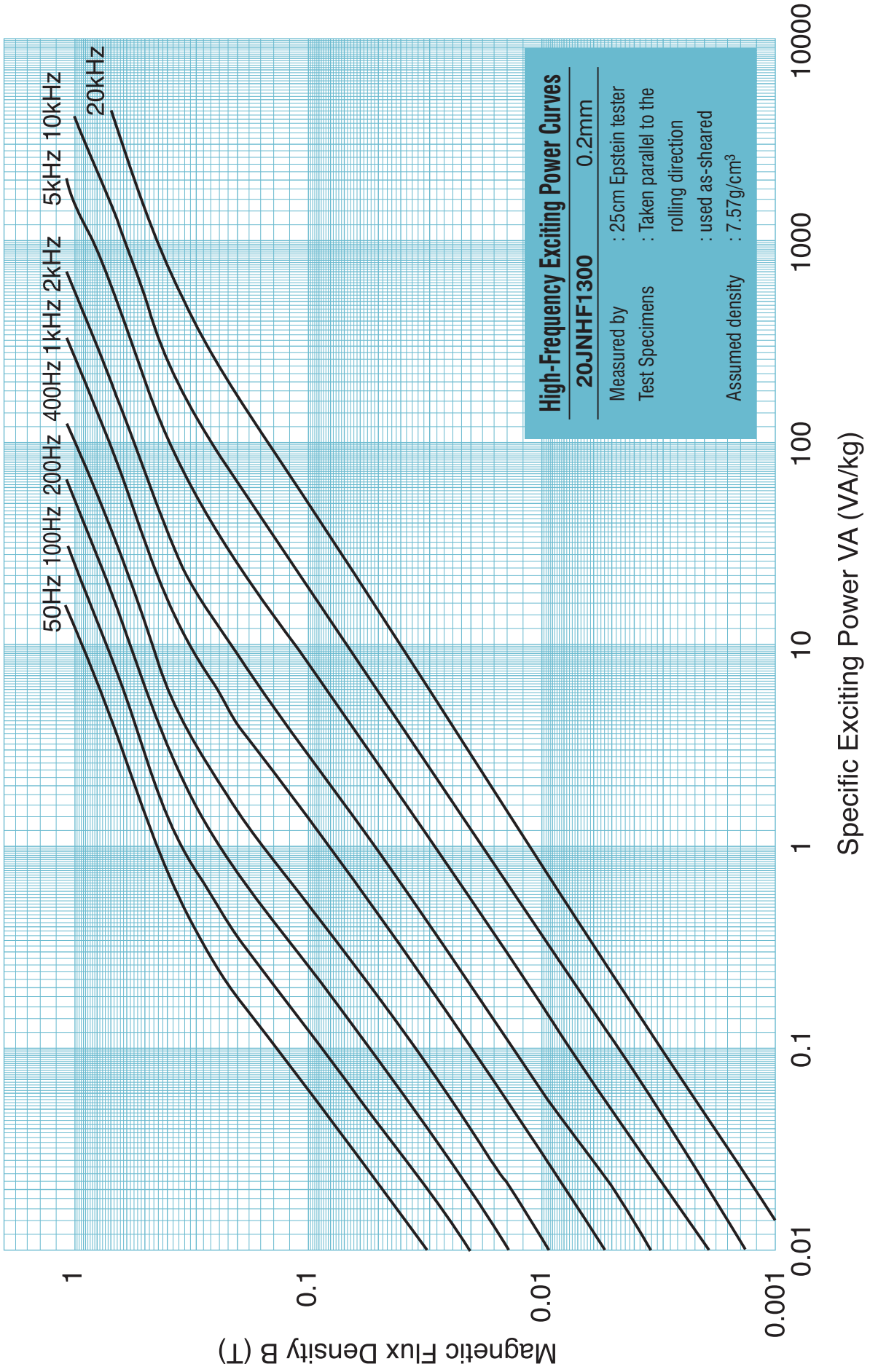

50Hz
100Hz
200Hz
400Hz
1kHz
2kHz
5kHz
10kHz
20kHz
Magnetic Flux Density B (T)
1
0.1
0.01
0.001
0.01
0.1
1
10
100
1000
Core Loss W (W/kg)
High-Frequency Core Loss Curves
20JNHF1300 0.2mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.57g/cm3


High-Frequency Magnetization Curves
20JNHF1300 0.2mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.57g/cm³
100Hz
2kHz
5kHz
10kHz
20kHz
Magnetic Flux Density B (T)
1
0.1
0.01
10
100
1000
Magnetizing Force H (A/m)



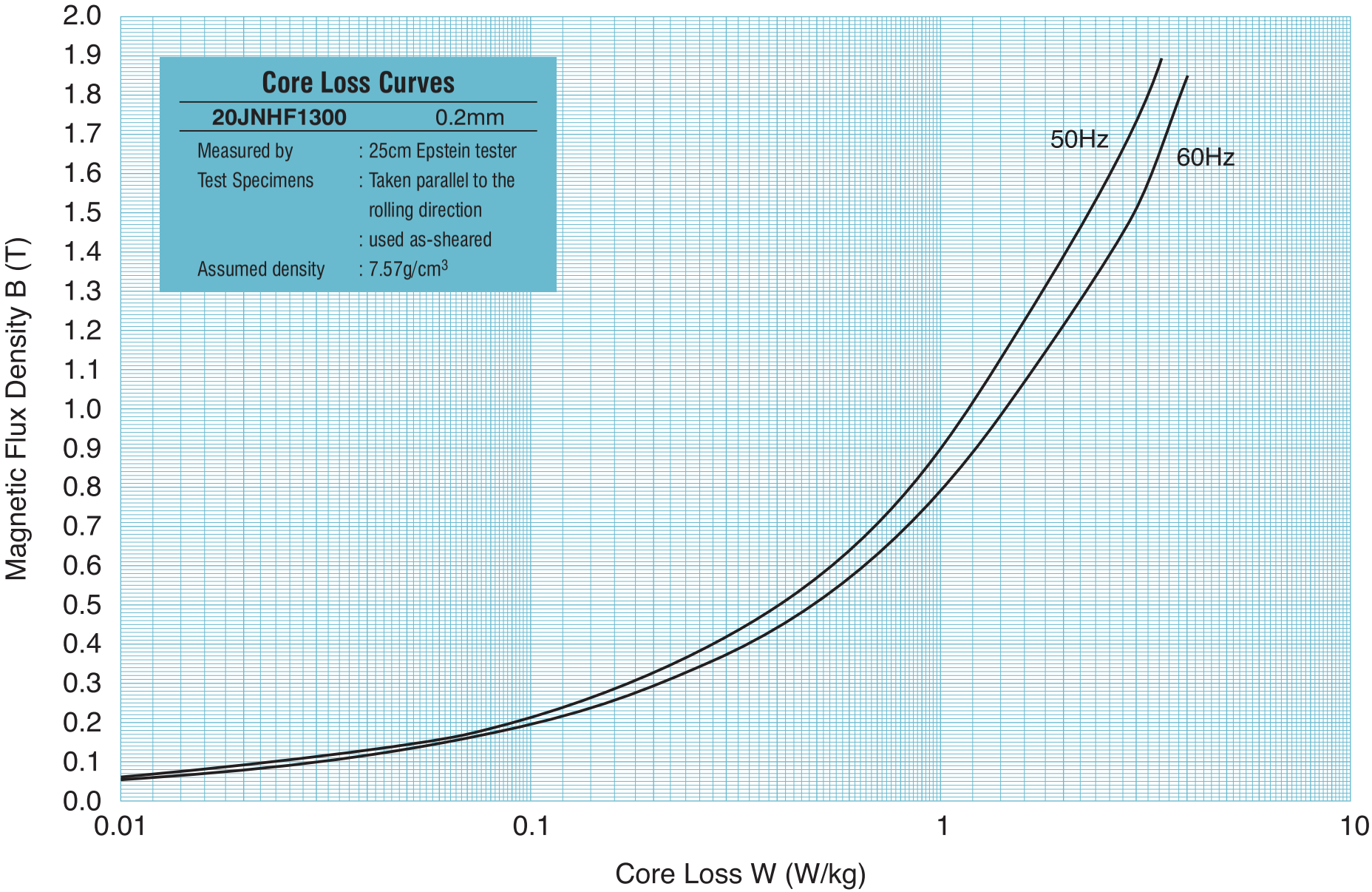
High-Frequency Excitation Curves

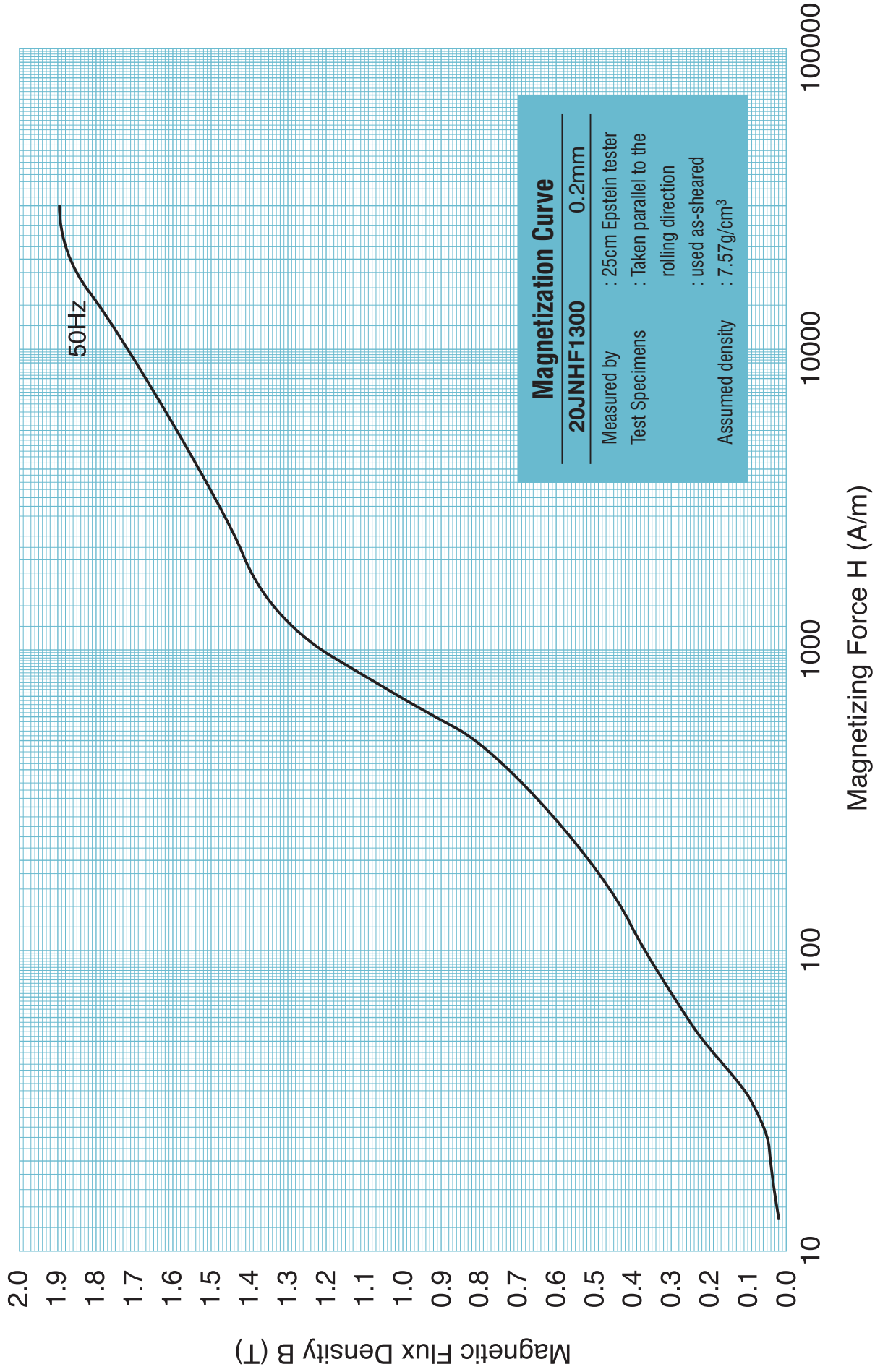
20JNHF1300	0.2mm
Measured by	: 25cm Epstein tester
Test Specimens	: Taken parallel to the rolling direction
	: used as-sheared
Assumed density	: 7.57g/cm^3

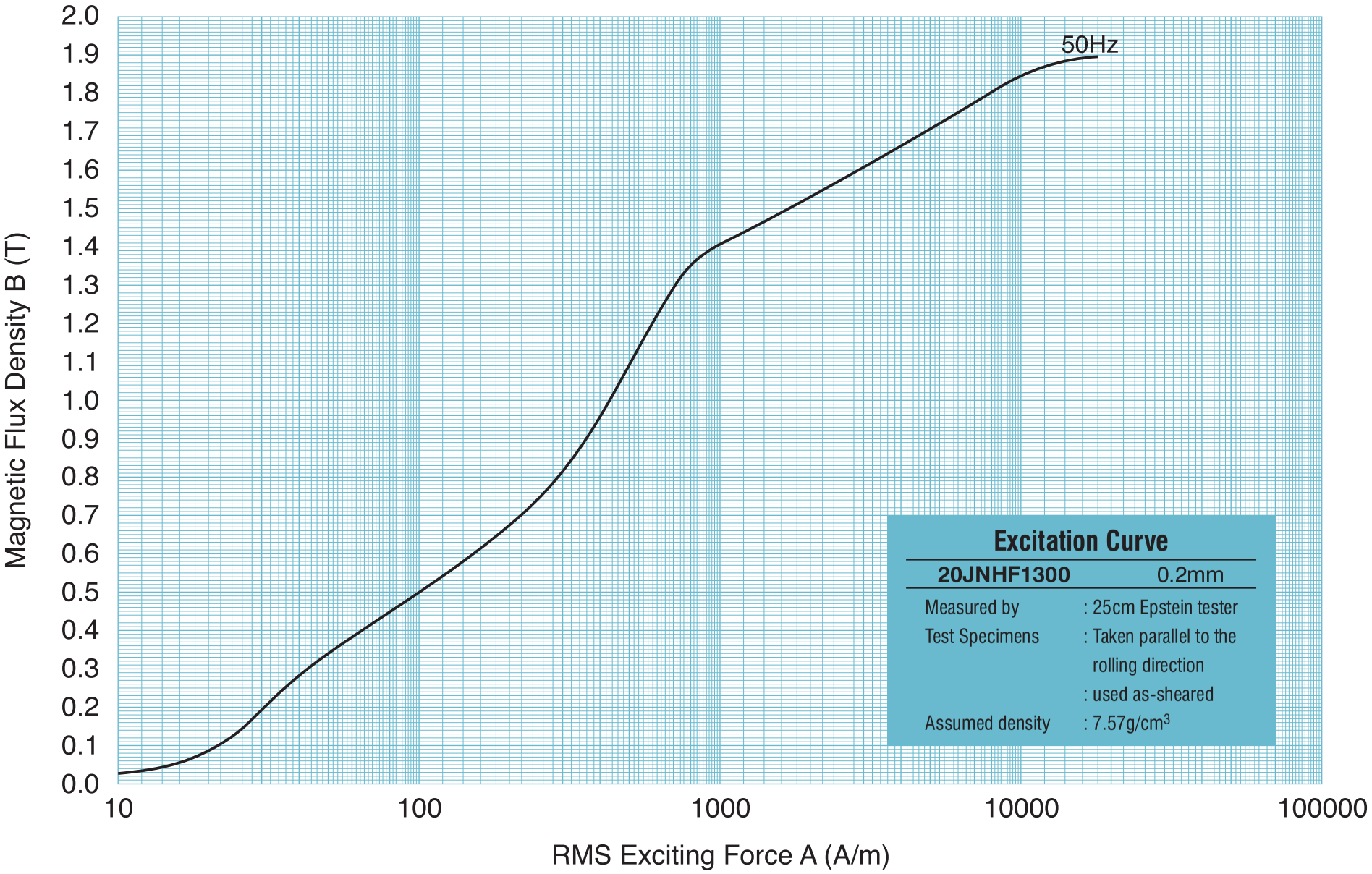

High-Frequency Exciting Power Curves
20JNHF1300 0.2mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.57g/cm3
50Hz
100Hz
200Hz
400Hz
1kHz
2kHz
5kHz
10kHz
20kHz
Magnetic Flux Density B (T)
1
0.1
0.01
0.001
Specific Exciting Power VA (VA/kg)
0.01
0.1
1
10
100
1000
10000

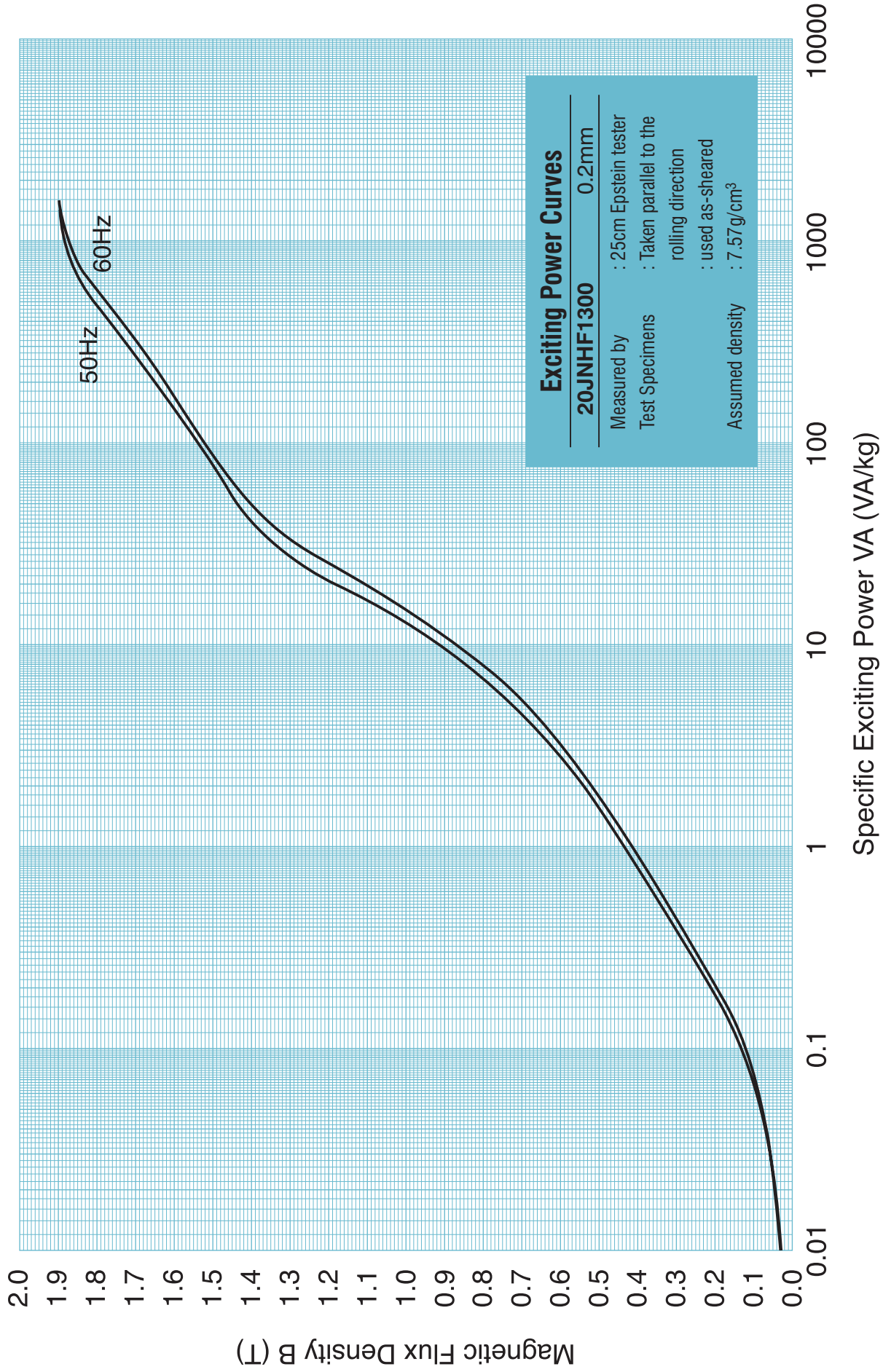
Core Loss Curves

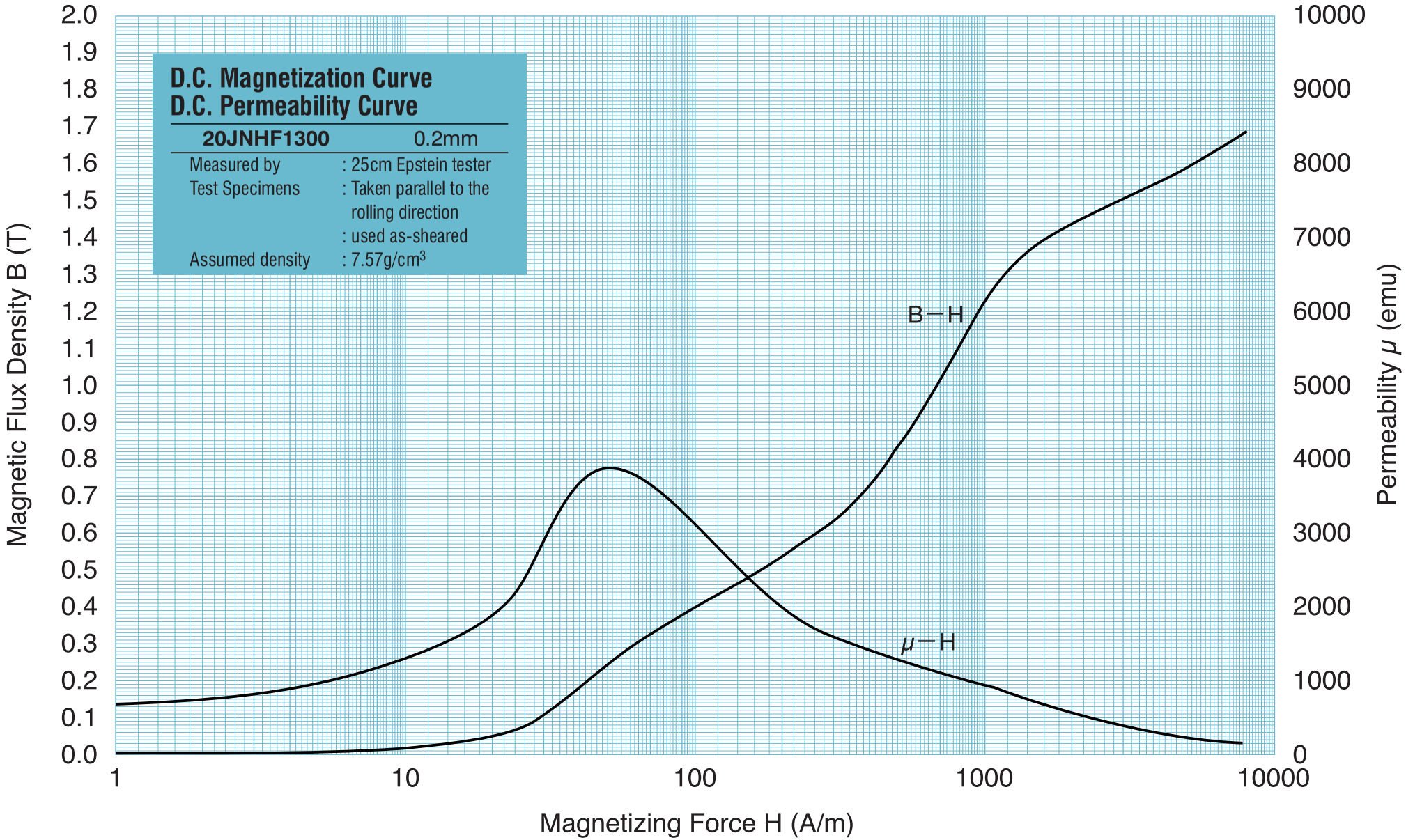
20JNHF1300	0.2mm
Measured by	: 25cm Epstein tester
Test Specimens	: Taken parallel to the rolling direction
	: used as-sheared
Assumed density	: 7.57g/cm^3

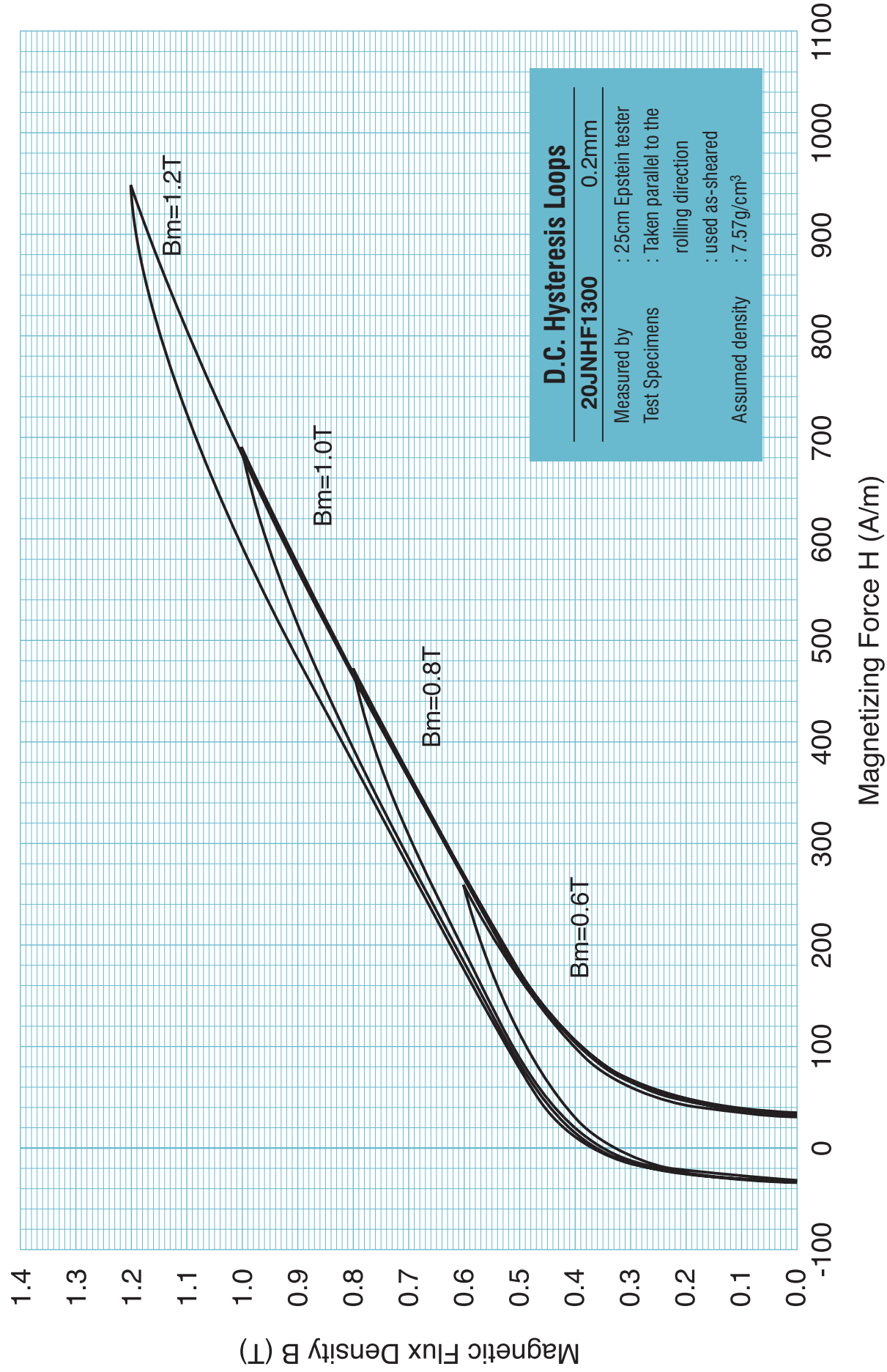



Magnetic Flux Density B (T)
2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
10
100
1000
10000
100000
Magnetizing Force H (A/m)
50Hz
Magnetization Curve
20JNHF1300 0.2mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.57g/cm3


2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Magnetic Flux Density B (T)
50Hz
10
100
1000
10000
100000
RMS Exciting Force A (A/m)
Excitation Curve
20JNHF1300
0.2mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.57g/cm3


Exciting Power Curves
20JNHF1300 0.2mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.57g/cm³
50Hz
60Hz
Magnetic Flux Density B (T)
Specific Exciting Power VA (VA/kg)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
1.1
1.2
1.3
1.4
1.5
1.6
1.7
1.8
1.9
2.0
0.01
0.1
1
10
100
1000
10000


D.C. Magnetization Curve
D.C. Permeability Curve
20JNHF1300 0.2mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.57g/cm3
B−H
μ−H
Magnetic Flux Density B (T)
Permeability μ (emu)
Magnetizing Force H (A/m)
2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
10000
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
1
10
100
1000
10000


D.C. Hysteresis Loops
20JNHF1300 0.2mm
Measured by : 25cm Epstein tester
Test Specimens : Taken parallel to the rolling direction
: used as-sheared
Assumed density : 7.57g/cm3
Bm=1.2T
Bm=1.0T
Bm=0.8T
Bm=0.6T
Magnetic Flux Density B (T)
Magnetizing Force H (A/m)
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
-100
0
100
200
300
400
500
600
700
800
900
1000
1100

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